

Xalqaro Fizika Olimpiadasi masalalari

Mexanika – I masala (8 ball)

Sakrovchi zarra

Zarra Ox musbat o'qi bo'ylab (bir o'lchovli holat) harakatlanadi. Unga ta'sir etuvchi kuchning Ox o'qidagi proyeksiyasi $F_x = F_0$ bo'lib, u x ga bog'liq holda quyidagi rasmda tasvirlangan. Ox o'qining boshida mukammal qaytaruvchi devor joylashgan. Zarra har qayerda bo'lmasin, unga moduli o'zgarmas $F_f = 1.00\text{ N}$ ishqalanish kuchi ta'sir qiladi. Zarra $x = x_0 = 1.00\text{ m}$ nuqtadan $E_0 = 10.0\text{ J}$ kinetik energiya bilan harakatni boshlaydi.

a. Zarra to'liq to'xtagunga qadar bosib o'tgan yo'l uzunligini toping. b. Zarraning F_x kuch maydonidagi potensial energiyasi $U(x)$ ning grafigini chizing. c. Zarra tezligining uning x koordinatasiga bog'liqligini sifat jihatidan chizing.

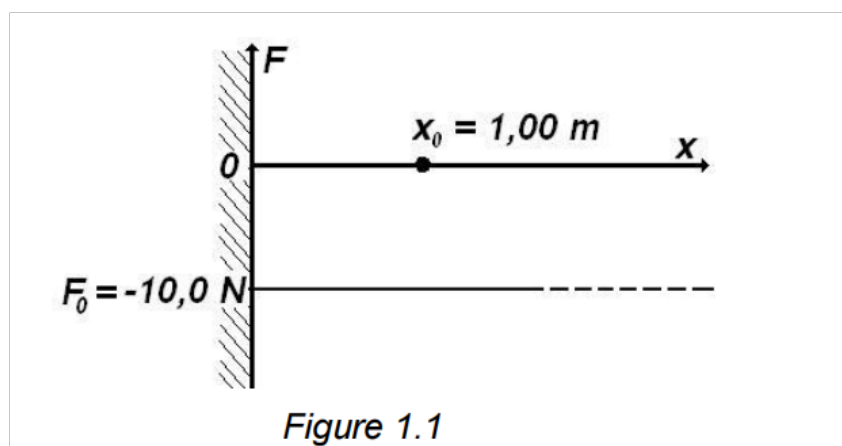


Figure 1: 1.1-rasm

I masala – Yechim

a. Masaladagi holatni modellashtirish mumkin: Ox o'qini vertikal yo'naltirilgan va pastki qismida devor joylashgan deb qaraymiz. Konservativ kuch F_x zarra og'irligi bo'lishi mumkin. Zarra harakatini kichik elastik sharchaning yerga elastik urilib, o'zgarmas ishqalanishli muhitda vertikal harakatiga o'xshatish mumkin. Ishqalanish kuchi og'irlikdan kichik. Zarraning potensial energiyasini sharchaning gravitatsion potensial energiyasi $m \cdot g \cdot h$ ga o'xshatib ifodalash mumkin, bunda $m \cdot g = |F_x|$; $h = x$. Ma'lumki, konservativ kuch maydonida potensial energiyaning o'zgarishi faqat zarra boshlang'ich va oxirgi vaziyatlariga bog'liq bo'lib, bu vaziyatlar orasidagi yo'lga bog'liq emas.

Masaladagi holatda, zarra devor tomon harakatlanganda unga ta'sir qiluvchi kuch devor tomonga yo'nalgan va moduli

$$F_f^+ = |F_x| - F_f \quad (1.1)$$

$$F_f^+ = 9 \text{ N}$$

$$F_f^- = 9 \text{ N} \quad (1.2)$$

Shuning uchun, zarra devor tomon harakati o'zgarimas tezlanishli harakat bo'lib, tezlanish moduli

$$a_{\leftarrow} = \frac{|F_x| - F_f}{m} \quad (1.3)$$

Harakat davomida zarra tezligi ortadi.

Devor bilan to'qnashganda, zarra oldingi tezlik moduliga teng, ammo qarama-qarshi yo'nalgan tezlik bilan orqaga harakatlanadi. Zarra devordan uzoqlashib, Ox o'qining musbat yo'nalishida harakatlanganda, unga ta'sir qiluvchi kuch yana devor tomonga yo'nalgan va kattaligi

$$F_x^+ = |F_x| + F_f \quad (1.4)$$

$$F_x^+ = 11 \text{ N}$$

$$F_x^+ = 11 \text{ N} \quad (1.5)$$

Mos ravishda, zarra devordan uzoqlashgan sari sekinlashadi va tezlanish moduli

$$a_{\rightarrow} = \frac{|F_x| + F_f}{m} \quad (1.6)$$

Bu harakat davomida zarra tezligi nolgacha kamayadi. Harakat davomida zarraga kuch ta'sir qilgani uchun, jism o'qning hech bir nuqtasida muvozanat holatiga ega bo'la olmaydi – faqat koordinata boshida potensial energiya nolga teng bo'lgani uchun istisno. Zarra faqat shu nuqtada to'liq to'xtashi mumkin. Konservativ kuchning $x_0 = 0$ nuqtadan x nuqttagacha bajargan ishi $L_{0 \rightarrow x}$ zarra potensial energiyasining o'zgarishi $U(x) - U(0)$ bilan quyidagicha bog'langan:

$$\begin{aligned} U(x) - U(0) &= -L_{0 \rightarrow x} \\ U(x) - U(0) &= - \int_0^x F_x dx = \int_0^x |F_x| dx = |F_x| \cdot x \end{aligned}$$

Agar $x = 0$ da zarra potensial energiyasi nolga teng deb qabul qilinsa, zarra konservativ kuch maydonidagi boshlang'ich potensial energiyasi

$$F(x) = F_0 \quad (1.8)$$

ko'rinishida bo'lib,

$$U(x_0) = |F_x| \cdot x_0 \quad (1.9)$$

Zarraning boshlang'ich kinetik energiyasi $E(x_0)$ berilganidek

$$E(x_0) = E_0 \quad (1.10)$$

va demak, zarra to'liq energiyasi $W(x_0)$

$$W(x_0) = U(x_0) + E_0 \quad (1.11)$$

Zarra to'liq to'xtashi uchun uning to'liq energiyasi ishqalanish kuchi ishi hisobiga butunlay sarflanishi kerak. Zarra to'xtaguncha bosgan masofa D quyidagi tenglamaga bo'ysunadi:

$$\begin{aligned} W(x_0) &= D \cdot F_f \\ U(x_0) + E_0 &= D \cdot F_f \\ |F_x| \cdot x_0 + E_0 &= D \cdot F_f \end{aligned} \quad (1.12)$$

Shunday qilib,

Agar harakat ishqalanishsiz sodir bo'lsa, zarra koordinata boshidan δ masofada joylashgan A nuqtaga yetib boradi, bu nuqtada kinetik energiya nolga teng bo'ladi. A nuqtada zarra energiyasi butunlay potensial energiyadan iborat.

Boshlang'ich nuqta va A nuqta uchun energiyaning saqlanish qonuni quyidagini beradi:

$$E_c + |F_x| \cdot x_0 = |F_x| \cdot \delta$$

$$\delta = x_0 + \frac{E_c}{|F_x|}$$

(1.16)

A nuqtaning koordinata boshidan eng uzoq masofadagi son qiymati, agar harakat ishqalanishsiz bo'lsa:

$$\delta = 2 \text{ m}$$

Potensial energiyaning vaziyatga bog'liqligi $(0, \delta)$ sohada 1.2-rasmda tasvirlangan.

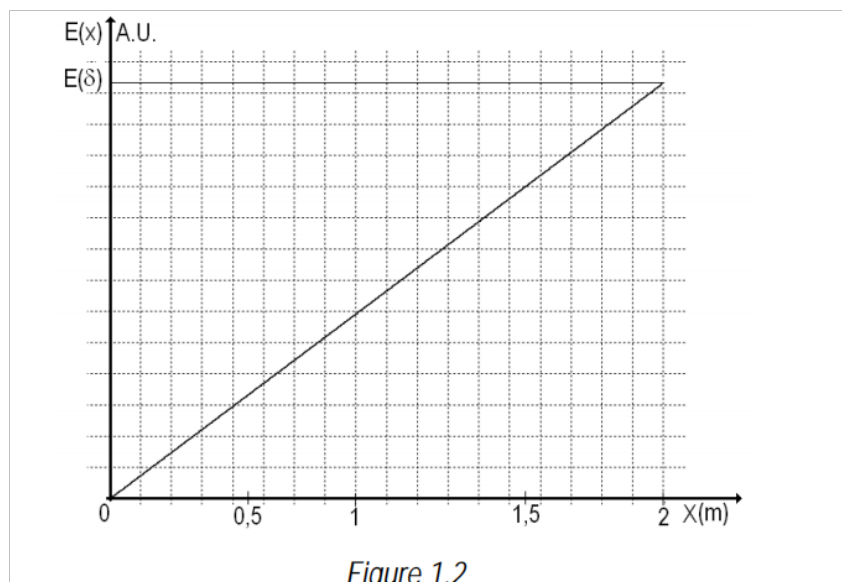


Figure 2: 1.2-rasm

Zarraning haqiqiy harakatida (ishqalanish bilan) erishiladigan eng chekka vaziyatlar δ dan kichik bo'ladi (chunki ishqalanish tufayli energiya yo'qoladi).

1.2-rasmdagi grafik I.b savolga javobdir.

c.

Zarra harakati davomida uning energiyasi ishqalanish kuchi bajargan dissipativ ish hisobiga kamayadi. Zarra tezligi devor yaqinida mahalliy maksimumga ega. v_k deb zarra k -chi marta devor bilan to'qnashuv oldidan tezligini, v_{k+1} esa keyingi to'qnashuv oldidan tezligini belgilasak, $v_k > v_{k+1}$. Ikki ketma-ket to'qnashuv orasida zarra tezligi nolga aylanadigan x_k vaziyatlarga yetadi va bu vaziyatlarda zarra energiyasi butunlay potensial energiyadan iborat bo'ladi. Bu vaziyatlar devorga tobora yaqinlashadi, chunki ishqalanish tufayli energianing bir qismi yo'qoladi. $x_{k+1} < x_k$

(1.17)

1-holat

Zarra devor tomon harakat qilganda uning tezligi ham, kinetik energiyasi ham ortadi. Potensial energiyasi esa kamayadi. Harakat davomida – yo'nalishidan qat'i nazar – ishqalanish kuchi hisobiga energiya yo'qoladi.

Harakatning bu qismida zarra potensial energiyasi $U(x)$, kinetik energiyasi $E(x)$ va to'liq energiyasi $W(x)$ quyidagi munosabatga bo'ysunadi:

$$W(x_0) - W(x) = F_f \cdot (x_0 - x)$$

(1.18)

bunda x vaziyati quyidagi sohaga tegishli:

$$x \in (0, x_0)$$

(1.19)

va x_0 dan koordinata boshiga tomon qarab bosib o'tiladi. (1.18) munosabatni quyidagicha yozish mumkin:

$$[E_c + |F_x| \cdot x_0] - \left[\frac{m \cdot v^2}{2} + |F_x| \cdot x \right] = F_f(x_0 - x)$$

(1.20)

bundan

$$v^2 = \frac{2}{m} [E_c + |F_x| \cdot x_0 - |F_x| \cdot x - F_f \cdot (x_0 - x)]$$

$$v^2 = \frac{2}{m} [E_c + x_0(|F_x| - F_f) - x(|F_x| - F_f)]$$

(1.21)

va natijada

$$v = -\sqrt{\frac{2}{m} [E_c + x_0(|F_x| - F_f) - x(|F_x| - F_f)]}$$

(1.22)

Tezlik moduli oldidagi minus ishorasi zarra harakati koordinata o'qining manfiy yo'nalishida ekanini bildiradi.

Berilgan ma'lumotlardan foydalanib:

$$v^2 = \frac{2}{m}(19 - 9x)$$

$$v = -\sqrt{\frac{2}{m}(19 - 9x)}$$

(1.23)

Zarraning devor bilan birinchi to'qnashuv oldidan tezligi $v_{1\leftarrow}$ quyidagicha ifodalanadi:

$$v_{1\leftarrow} = -\sqrt{\frac{2}{m} [E_c + x_0(|F_x| - F_f)]}$$

(1.24)

va uning son qiymati

$$v_{1\leftarrow} = -\sqrt{\frac{2 \cdot 19}{m}}$$

(1.25)

Devor yaqinidagi to'liq energiya, butunlay kinetik energiya $E_{1\leftarrow}$ quyidagicha:

$$E_{1\leftarrow} = E_c + x_0(|F_x| - F_f)$$

(1.26)

Bu energiyaning son qiymati

$$E_{1\leftarrow} = 19 \text{ J}$$

(1.27)

1.3-rasmdagi grafik zarra harakatining birinchi qismida tezlik kvadratining vaziyatga bogʻliqligini koʻrsatadi.

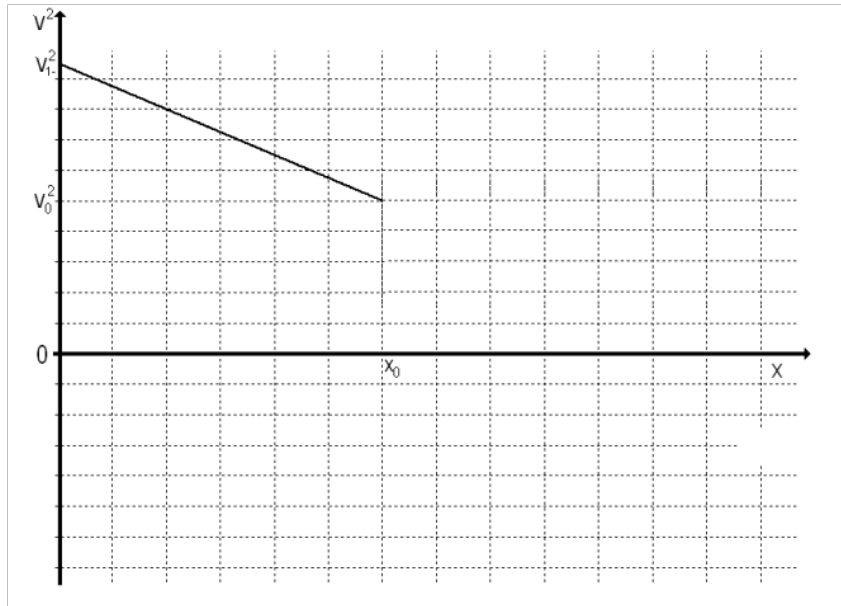


Figure 3: 1.3-rasm

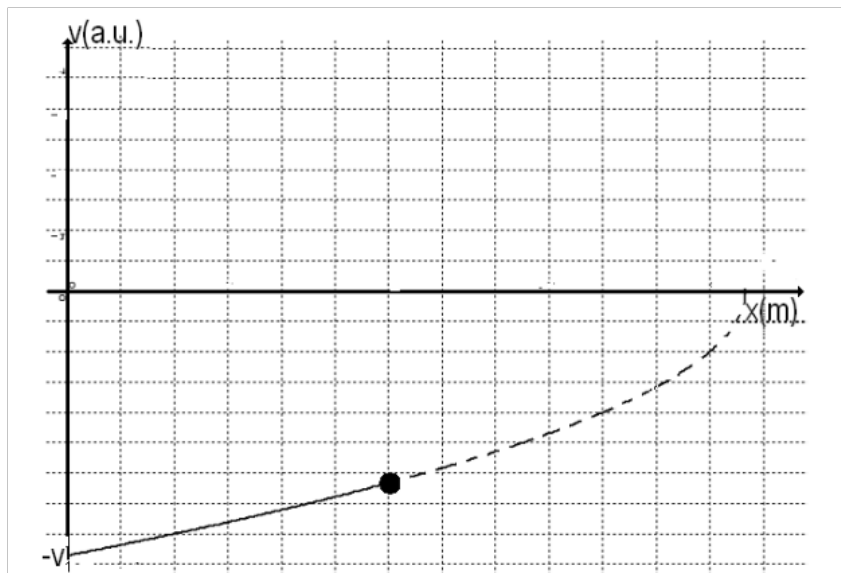


Figure 4: 1.4-rasm

1.4-rasmdagi grafik zarra harakatining birinchi qismida (devor tomon) tezlikning vaziyatga bogʻliqligini tasvirlaydi.

Devor bilan to'qnashgandan so'ng, zarra tezligi $v_{1\leftrightarrow}$ to'qnashuv oldidan tezlik moduliga teng, ammo yo'nalishi qarama-qarshi bo'ladi. Tezlikning vaziyat funksiyasi sifatidagi grafigida devor bilan to'qnashuv tezlik o'qining manfiy tomonidan musbat tomoniga sakrash ko'rinishida aks etadi. To'qnashuv oldidan va to'qnashuvdan so'ng tezlikning absolyut qiymati bir xil bo'lib, 1.5-rasmda ko'rsatilgan.

$$v_{1\leftrightarrow} = \sqrt{\frac{2}{m} [E_c + x_0(|F_x| - F_f)]}$$

(1.28)

Birinchi to'qnashuvdan keyin zarra harakati o'zgarmas manfiy tezlanish a_x va boshlang'ich tezlik $v_{1\leftrightarrow}$ bilan sekinlashadi.

Bu harakat tezlik nolga aylanadigan x_1 vaziyatgacha davom etadi.

Galiley qonunidan:

$$0 = v_{1\leftrightarrow}^2 - 2 \cdot a_x \cdot x_1$$

$$x_1 = \frac{v_{1\leftrightarrow}^2}{2 \cdot a_x} = \frac{\frac{2}{m} [E_c + x_0(|F_x| - F_f)]}{2 \cdot \frac{|F_x| + F_f}{m}} = \frac{E_c + x_0(|F_x| - F_f)}{|F_x| + F_f}$$

(1.29)

x_1 vaziyatning son qiymati

$$x_1 = \frac{19}{11} \text{ m}$$

(1.30)

Koordinata boshidan x_1 ga tomon bosib o'tilgan vaziyatlar

$$x \in (0, x_1)$$

(1.31)

uchun to'liq energiya $W(x)$ quyidagicha:

$$W(x) = \frac{m \cdot v^2}{2} + |F_x| \cdot x$$

(1.32)

Devordan uzoqlashganda zarra energiyasi ishqalanish tufayli kamayadi, ya'ni

$$E_{1\leftrightarrow} - W(x) = F_f \cdot x$$

(1.33)

$$E_c + x_0(|F_x| - F_f) - \frac{m \cdot v^2}{2} - |F_x| \cdot x = F_f \cdot x$$

Tezlik modulining kvadrati:

$$v^2 = \frac{2}{m} [E_c + x_0(|F_x| - F_f) - (|F_x| + F_f) \cdot x]$$

(1.35)

Berilgan ma'lumotlardan foydalanib:

$$v^2 = \frac{2}{m}[19 - 11 \cdot x]$$

(1.36) va mos ravishda

$$V = \sqrt{\frac{2}{m}[19 - 11 \cdot x]}$$

(1.37)

$x \in (0, x_1)$ sohadagi vaziyatlar uchun (bu zarra harakatining ikkinchi qismiga to'g'ri keladi) 1.5-rasm tezlikning vaziyatga bog'liqligini ko'rsatadi.

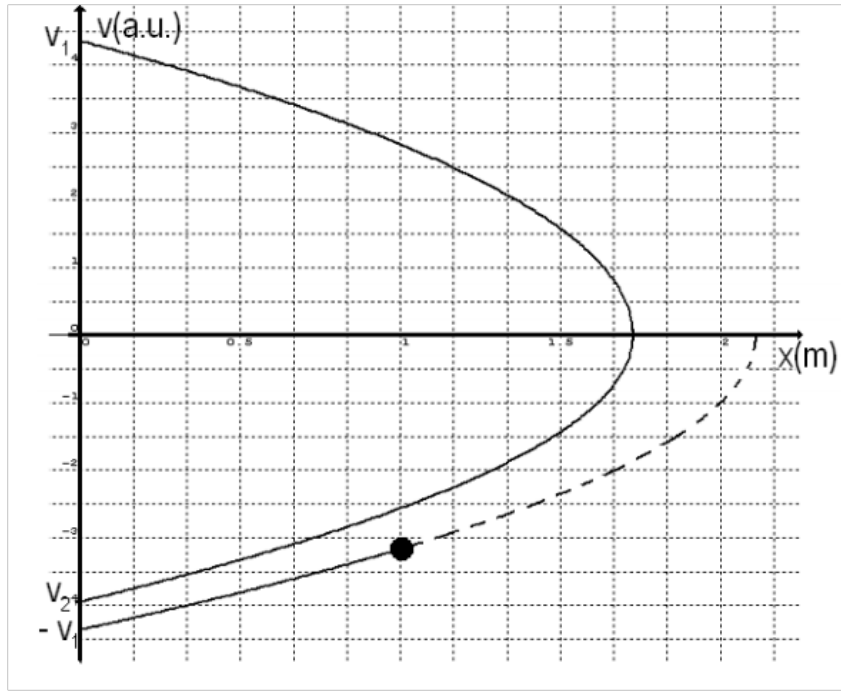


Figure 5: 1.5-rasm

Rasmdan ko'rinib turibdiki, zarra eng uzoq X_1 vaziyatga yetgandan so'ng, boshlang'ich tezliksiz, tezlanish moduli $a_x = (|F_x| - F_f)/m$ bo'lgan tezlanuvchan harakat bilan koordinata boshiga tomon harakatlanadi. Devor bilan to'qnashgandan keyin zarra to'qnashuv oldidan tezlik moduliga teng, ammo qarama-qarshi yo'nalgan tezlikka ega bo'ladi. Zarra x_1 dan koordinata boshiga tomon $(0, x_1)$ sohadagi biror nuqtaga yetganda uning to'liq energiyasi $W(x)$ (1.32) ifoda bilan beriladi. x_1 dan boshlab, ishqalanish kuchi tufayli sodir bo'ladigan dissipatsiya hisobiga energiya x koordinatali vaziyatga mos keladigan qiymatgacha o'zgaradi.

$$|F_x| \cdot x_1 - W(x) = F_f \cdot (x_1 - x)$$

$$|F_x| \cdot x_1 - \frac{m \cdot v^2}{2} - |F_x| \cdot x = F_f \cdot (x_1 - x)$$

(1.38) Tezlik kvadratining ifodasi:

$$v^2 = \frac{2}{m} [(|F_x| - F_f) \cdot (x_1 - x)]$$

$$v^2 = \frac{2}{m} [E_C + x_0(|F_x| - F_f) - (|F_x| + F_f) \cdot x]$$

(1.39) va tezlik

$$V = \sqrt{\frac{2}{m} [E_C + x_0(|F_x| - F_f) - (|F_x| + F_f) \cdot x]}$$

(1.40)

Berilgan ma'lumotlardan foydalanib, $(0, x_1)$ sohadagi vaziyat uchun:

$$v^2 = \frac{2}{m} \left[\frac{19}{11} - x \right] \cdot 9$$

(1.41) mos ravishda

$$V = -\sqrt{\frac{2}{m} \left[\frac{19}{11} - x \right] \cdot 9}$$

(1.42)

Zarraning devorga ikkinchi marta yetib kelgandagi tezligi (1.39) dan foydalanib:

$$V_{2\leftarrow} = \sqrt{\frac{2}{m} [E_C + x_0(|F_x| - F_f)] - \frac{(|F_x| - F_f)}{(|F_x| + F_f)}}$$

(1.43) hosil bo'lgan son qiymat:

$$V_{2\leftarrow} = -\sqrt{\frac{2}{m} \frac{171}{11}}$$

(1.44)

Xulosa qilib aytganda, birinchi to'qnashuv va birinchi qaytishdan so'ng zarra devordan uzoqlashadi, tezligi nolga aylanadigan vaziyatga yetadi va yana devorga qaytadi. Zarraning devorga qayta urilish tezligi oldingisidan kichik bo'ladi – 1.5-rasmdagidek. Yuqorida belgilanganidek, v_k zarra k -chi urilish oldidan tezligi, x_k esa k -chi urilishdan keyin erishilgan eng uzoq vaziyat koordinatasi. Devordan boshlanganda zarra energiyasi:

$$E_k = \frac{v_k^2 \cdot m}{2} = W_k(0)$$

(1.45)

k -chi urilishdan keyin koordinata boshidan eng uzoq x_k nuqtada energiya quyidagi munosabatni qanoatlantiradi:

$$U_k = x_k \cdot |F_x| = W_k(x_k)$$

(1.46)

Boshlang'ich nuqta va x_k nuqta orasidagi energiya o'zgarishi:

$$\frac{v^2 \cdot m}{2} - x_k \cdot |F_x| = F_f \cdot x_k$$

(1.47)

bundan

$$x_k = \frac{v^2 \cdot m}{2 \cdot (|F_x| + F_f)}$$

(1.48)

$$v_{(k,\leftarrow)}^2 = \frac{2}{m} \left[9 \cdot \left(\frac{19}{11} \right)^k - x \right]$$

(1.58)

Koordinata boshidan boshlanadigan k -chi urilish uchun (1.34) ga o'xshash, tezlik moduli kvadratining vaziyatga bog'liqligi $v_{(k,\rightarrow)}^2$ quyidagicha yoziladi:

$$\begin{aligned} v_{(k,\rightarrow)}^2 &= \frac{2}{m} [(|F_x| + F_f) \cdot (X_k - x)] \\ v_{(k,\rightarrow)}^2 &= \frac{2}{m} [(|F_x| + F_f) \cdot (x_k \cdot q^{2k} - x)] \end{aligned}$$

(1.59)

Berilgan ma'lumotlardan foydalanib:

$$v_{(k,\rightarrow)}^2 = \frac{2}{m} \left[11 \cdot \left(\frac{19}{11} \right)^k - x \right]$$

(1.60)

Tezlik kvadratining vaziyatga bog'liq holda o'zgarishi 1.6-rasmda tasvirlangan.

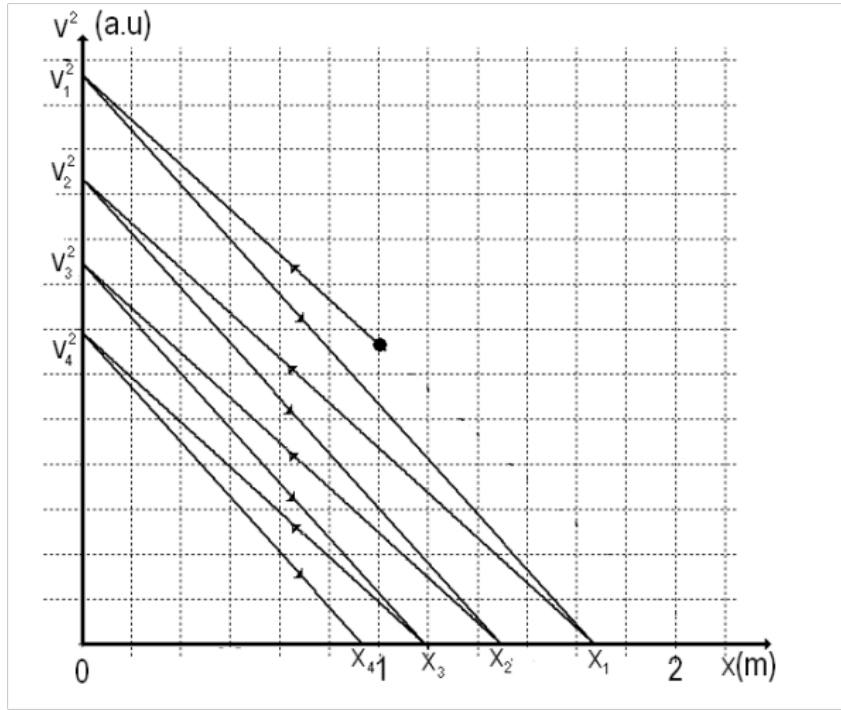


Figure 6: 1.6-rasm

Tezlikning vaziyatga bog‘liq holda o‘zgarishi esa 1.7-rasmda tasvirlangan.

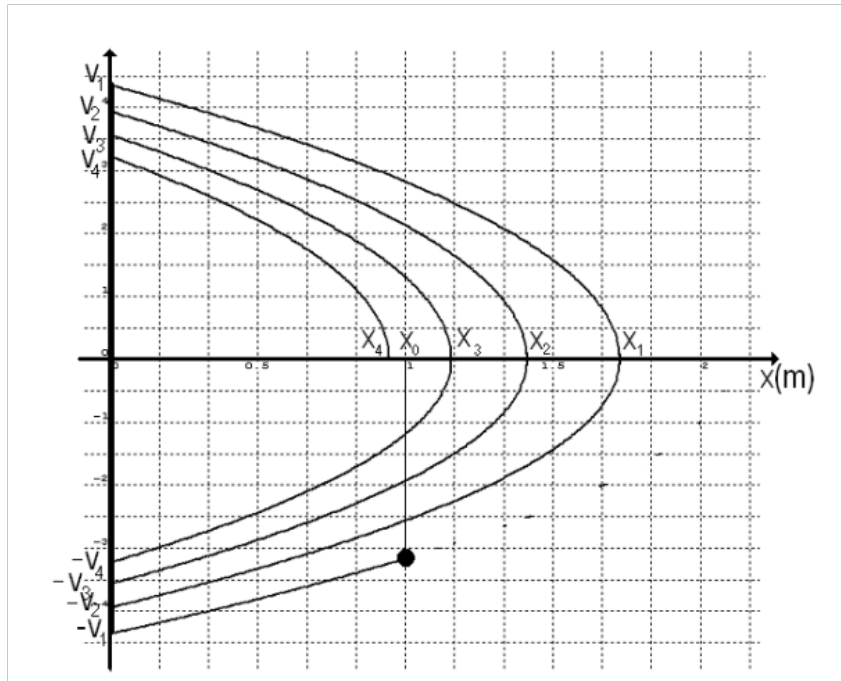


Figure 7: 1.7-rasm

(1.56) da berilgan progressiya yig‘indisi zarra birinchi urilishdan keyin bosib o‘tgan masofaning yarmini beradi.

$$\sum_{k=1}^{\infty} X_k = X_1 \frac{1}{1 - q^2}$$

(1.61)

(1.53) va (1.29) ni hisobga olgan holda:

$$\sum_{k=1}^{\infty} X_k = \frac{E_c + x_0(|F_x| - F_f)}{2 \cdot F_f}$$

(1.62)

Son jihatdan:

$$\sum_{k=1}^{\infty} X_k = \frac{19}{2} \text{ m}$$

(1.63)

Umumiy bosib o'tilgan masofa:

$$D = 2 \cdot \sum_{k=1}^{\infty} X_k + X_0$$

(1.64) $D = 20 \text{ m}$

bu (1.14) bilan bir xil.

2-holat

Agar zarra x_0 vaziyatdan Ox o'qining musbat yo'nalishi bo'ylab harakatni boshlasa, uning tezligi va kinetik energiyasi kamayadi, potensial energiyasi esa x'_1 vaziyatda tezlik nolga aylanadigan maksimumgacha ortadi. Bu harakat davomida ishqalanish tufayli energiya yo'qoladi.

x_0 va x'_1 orasidagi x vaziyatlar uchun to'liq energiya $W(x)$ quyidagi munosabatni qanoatlantiradi:

$$W(x_0) - W(x) = F_f(x - x_0) \quad (1.65)$$

bunda x vaziyati

$$x \in (x_0, x'_1) \quad (1.66)$$

sohaga tegishli bo'lib, zarra x_0 dan o'qning musbat yo'nalishi bo'ylab harakatlanadi.

(1.65) munosabat quyidagicha yoziladi:

$$[E_c + |F_x| \cdot x_0] - \left[\frac{m \cdot v^2}{2} + |F_x| \cdot x \right] = F_f(x - x_0)$$

(1.67)

bundan

$$v^2 = \frac{2}{m} [E_c + |F_x| \cdot x_0 - |F_x| \cdot x - F_f \cdot (x - x_0)]$$

va

$$v^2 = \frac{2}{m} [E_c + x_0(|F_x| + F_f) - x(|F_x| + F_f)]$$

(1.68)

$$v = \sqrt{\frac{2}{m}[E_c + x_0(|F_x| + F_f) - x(|F_x| + F_f)]}$$

(1.69)

Berilgan ma'lumotlardan foydalanib:

$$v^2 = \frac{2}{m}(21 - 11 \cdot x)$$

$$v = \sqrt{\frac{2}{m}(21 - 11 \cdot x)}$$

(1.70)

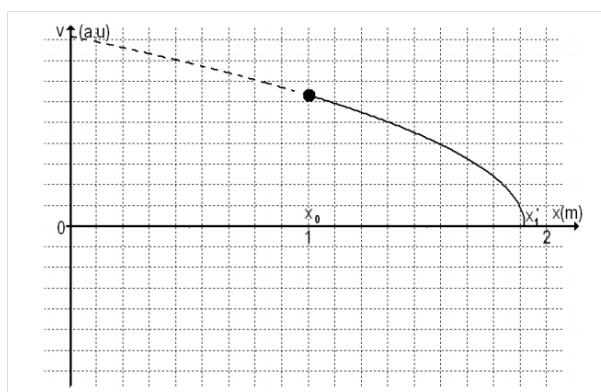
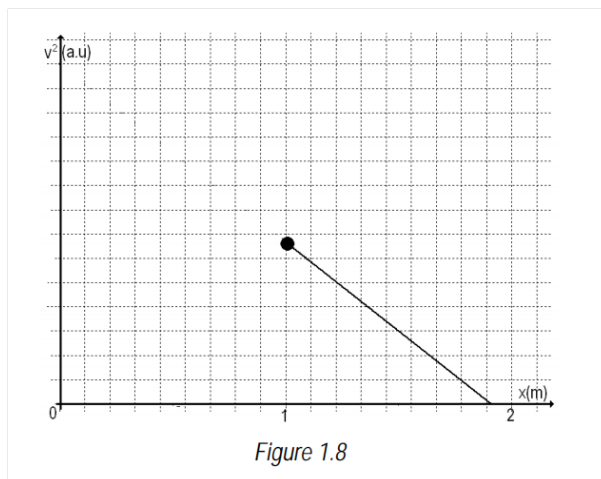


Figure 8: 1.9-rasm

1.8-rasmdagi grafik $x \in (x_0, x'_1)$ sohada harakat uchun tezlik kvadratining vaziyatga bog'liqligini ko'rsatadi. Zarra Ox o'qining musbat yo'nalishi bo'ylab harakatlanadi. Bu harakat tezlik nolga aylanadigan x'_1 vaziyatga yetguncha davom etadi. (1.68) munosabatda tezlik modulini nolga tenglab:

$$x'_1 = x_0 + \frac{E_c}{|F_x| + F_f}$$

(1.71)

x'_1 ning son qiymati:

$$x'_1 = \frac{21}{11} \text{ m} \quad (1.72)$$

Eng uzoq x'_1 vaziyatga yetgandan so'ng, zarra boshlang'ich tezliksiz, tezlanish moduli $a_{\leftarrow} = (|F_x| - F_f)/m$ bo'lgan tezlanuvchan harakat bilan yana koordinata boshiga tomon harakatlanadi. Devor bilan to'qnashgandan keyin zarra to'qnashuv oldidan tezlik moduliga teng, ammo qarama-qarshi yo'nalgan $v'_{1\leftarrow}$ tezlikka ega bo'ladi.

Zarra x'_1 dan koordinata boshiga tomon $(0, x'_1)$ sohadagi biror nuqtada bo'lganda uning to'liq energiyasi $W(x)$ quyidagicha:

$$W(x) = \frac{m \cdot v^2}{2} + |F_x| \cdot x$$

(1.73)

Ishqalanish tufayli energiya qiymati x'_1 dagi qiymatdan x vaziyatga mos keladigan qiymatgacha kamayadi:

$$|F_x| \cdot x'_1 - W(x) = F_f \cdot (x'_1 - x)$$

$$|F_x| \cdot x'_1 - \frac{m \cdot v^2}{2} - |F_x| \cdot x = F_f \cdot (x'_1 - x)$$

(1.74)

Tezlik kvadratining ifodasi:

$$v^2 = \frac{2}{m} [(|F_x| - F_f) \cdot (x'_1 - x)]$$

(1.75)

va tezlik:

$$v = -\sqrt{\frac{2}{m} [(|F_x| - F_f) \cdot (x'_1 - x)]}$$

(1.76)

Berilgan ma'lumotlar uchun $(0, x'_1)$ sohada:

$$v^2 = \frac{2}{m} \left[\frac{21}{11} - x \right] \cdot 9$$

(1.77)

mos ravishda

$$v = -\sqrt{\frac{2}{m} \left[\frac{21}{11} - x \right] \cdot 9}$$

(1.78)

Zarraning devorga ikkinchi marta urilish tezligi (1.78) ga asosan:

$$v'_{1\leftarrow} = -\sqrt{\frac{2}{m}[(|F_x| - F_f) \cdot x'_1]}$$

(1.79)

va uning qiymati:

$$v'_{1\leftarrow} = -\sqrt{\frac{2}{m} \frac{189}{11}}$$

(1.80)

Xulosa qilib aytganda, birinchi to'qnashuv va birinchi qaytishdan so'ng zarra devordan uzoqlashadi, tezligi nolga aylanadigan vaziyatga yetadi va yana devorga qaytadi. Zarraning devorga qayta urilish tezligi oldingisidan kichik bo'ladi – 1.11-rasmdagidek.

v'_k devordan k -chi urilish boshlanishidagi tezlik, x'_k esa k -chi urilishdagi eng uzoq vaziyat koordinatasi deb belgilasak, zarra devordan chiqish energiyasi:

$$\frac{v'^2_k \cdot m}{2} = E'_k = W'_k(0) \quad (1.81)$$

k -chi devordan uzoqlashishdan keyin x'_k vaziyatda energiya:

$$U'_k = x'_k |F_x| = W'_k(x'_k) \quad (1.82)$$

To'liq energiyaning o'zgarishi:

$$\frac{v'^2_k \cdot m}{2} - x'_k |F_x| = F_f \cdot x'_k \quad (1.83)$$

bundan

$$x'_k = \frac{v'^2_k \cdot m}{2(|F_x| + F_f)} \quad (1.84)$$

Zarra x'_k vaziyatga yetgandan so'ng tezlik yo'nalishi o'zgaradi va devorga urilganda:

$$\frac{v'^2_{k+1} \cdot m}{2} = E'_{k+1} = W'_{k+1}(0) \quad (1.85)$$

x'_k vaziyat va devorga urilish nuqtasi uchun energiyaning saqlanish qonuni:

$$x'_k |F_x| - \frac{v'^2_{k+1} \cdot m}{2} = F_f \cdot x'_k \quad (1.86)$$

bundan

$$\frac{v'^2_{k+1} \cdot m}{2} = x'_k (|F_x| - F_f) \quad (1.87)$$

(1.84) ni hisobga olgan holda (1.87) quyidagicha yoziladi:

$$v_{k+1}'^2 = v_k'^2 \cdot \frac{|F_x| - F_f}{|F_x| + F_f} \quad (1.88)$$

Ikki ketma-ket urilish orasida tezlik q maxrajli geometrik progressiya bo'yicha kamayadi. Bu maxraj quyidagiga teng:

$$q = \sqrt{\frac{|F_x| - F_f}{|F_x| + F_f}} \quad (1.89)$$

Berilgan ma'lumotlardan foydalanib:

$$q = \sqrt{\frac{9}{11}} \quad (1.90)$$

$(k+1)$ -chi urilish uchun (1.84) munosabat:

$$x_{k+1}' = \frac{v_{k+1}'^2 \cdot m}{2(|F_x| + F_f)} \quad (1.91)$$

(1.84) va (1.91) ni hisobga olgan holda, ikki ketma-ket urilishdagi eng uzoq vaziyatlar nisbati:

$$\frac{x_{k+1}'}{x_k'} = \frac{|F_x| - F_f}{|F_x| + F_f} = q^2 \quad (1.92)$$

$$x_{k+1}' = q^2 x_k'$$

Koordinata boshiga tomon k -chi harakat uchun (1.57) ga o'xshash, tezlik kvadratining vaziyatga bog'liqligi $v_{(k,\leftarrow)}^2$ quyidagicha yoziladi:

$$\begin{aligned} v_{(k,\leftarrow)}^2 &= \frac{2}{m} [(|F_x| - F_f) \cdot (x_k' - x)] \\ v_{(k,\leftarrow)}^2 &= \frac{2}{m} [(|F_x| - F_f) \cdot (x_1' q^{2k} - x)] \end{aligned} \quad (1.93)$$

Yoki berilgan ma'lumotlardan foydalanib:

$$v_{(k,\leftarrow)}^2 = \frac{2}{m} \left[9 \cdot \left(\frac{21}{11} \right) \cdot \left(\frac{9}{11} \right)^k - x \right] \quad (1.94)$$

Koordinata boshidan k -chi harakat uchun (1.59) ga o'xshash, tezlik kvadratining vaziyatga bog'liqligi $v_{(k,\rightarrow)}^2$ quyidagicha yoziladi:

$$\begin{aligned} v_{(k,\rightarrow)}^2 &= \frac{2}{m} [(|F_x| + F_f) \cdot (x_k' - x)] \\ v_{(k,\rightarrow)}^2 &= \frac{2}{m} [(|F_x| + F_f) \cdot (x_1' q^{2k} - x)] \end{aligned} \quad (1.95)$$

Berilgan ma'lumotlardan foydalanib:

$$v_{(k,\rightarrow)}^2 = \frac{2}{m} \left[11 \cdot \left(\frac{21}{11} \right) \cdot \left(\frac{9}{11} \right)^k - x \right] \quad (1.96)$$

Tezlik kvadratining vaziyatga bogʻliq holda oʻzgarishi 1.10-rasmda tasvirlangan.

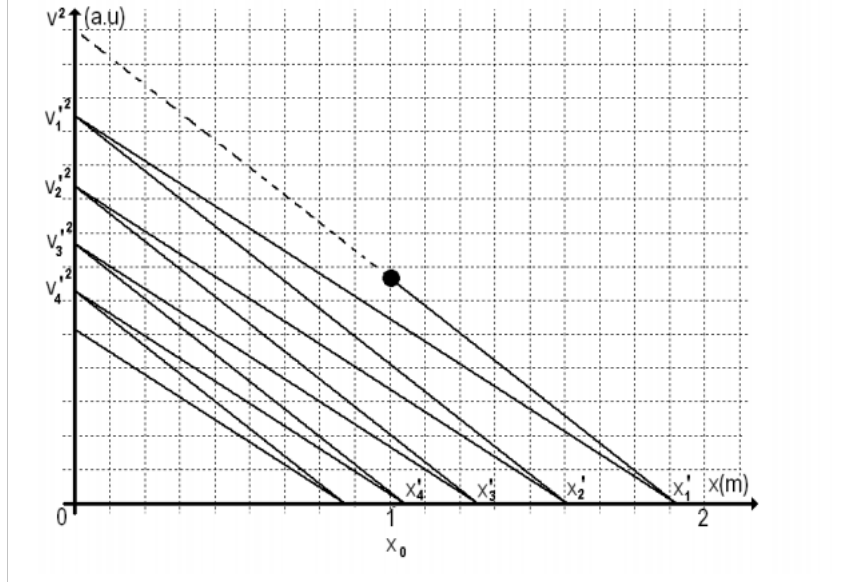


Figure 9: 1.10-rasm

Tezlikning vaziyatga bogʻliq holda oʻzgarishi esa 1.11-rasmda tasvirlangan.

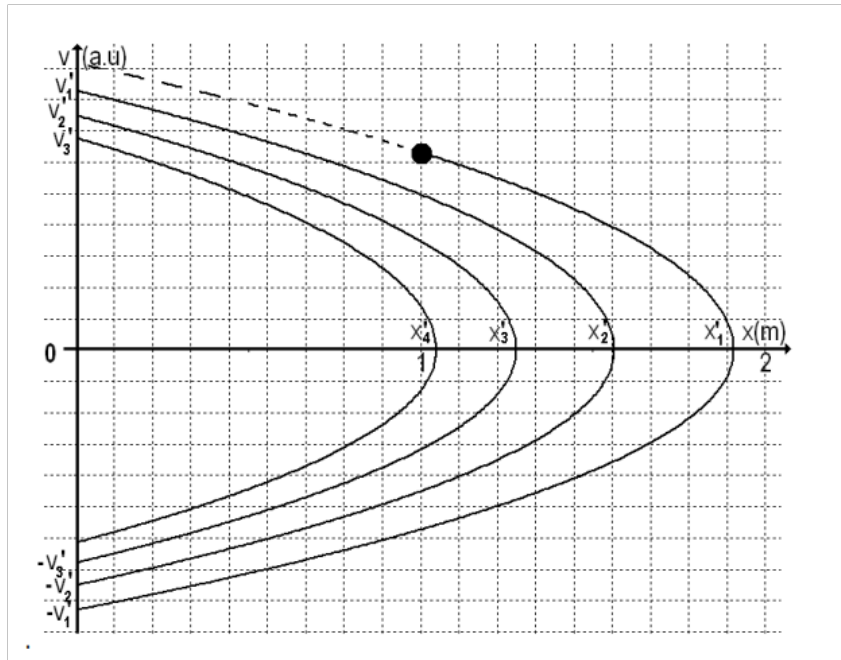


Figure 10: 1.11-rasm

(1.92) geometrik progressiya yigʻindisi (ikkiga koʻpaytirilib, x_0 ayrilganda) zarra bosib oʻtgan umumiy masofani beradi.

$$\sum_{k=1}^{\infty} x_k = x'_1 \frac{1}{1 - q^2} \quad (1.97)$$

(1.97), (1.71) va (1.72) ni hisobga olgan holda:

$$\sum_{k=1}^{\infty} x_k = \frac{21}{2} \text{ m} \quad (1.98)$$

Zarra bosib o'tgan umumiy masofa:

$$D = 2 \cdot \sum_{k=1}^{\infty} x_k - x_0 \quad (1.99)$$

$$D = 20 \text{ m}$$

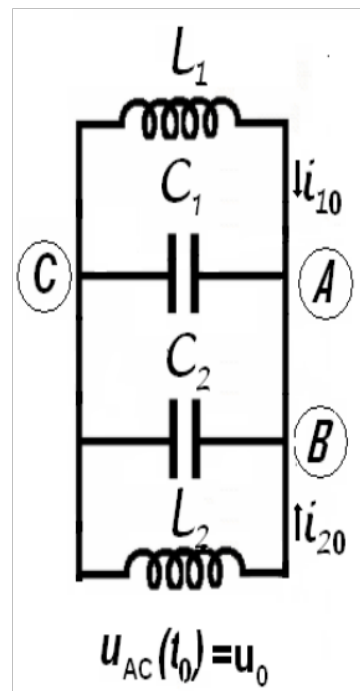
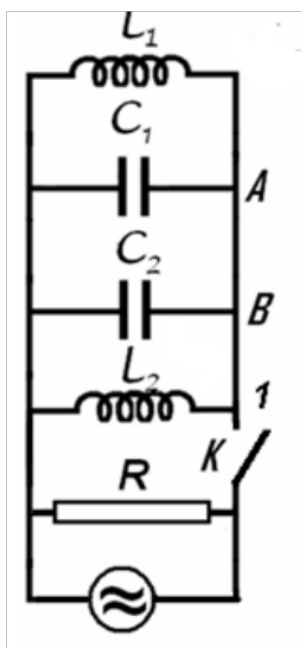
bu (1.14) natijani yana bir bor tasdiqlaydi.

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Elektr – II masala (8 ball)

Har xil tebranish

Rasmda keltirilgan elektr zanjirini ko'rib chiqamiz, bunda $L_1 = 10 \text{ mH}$, $L_2 = 20 \text{ mH}$, $C_1 = 10 \text{ nF}$, $C_2 = 5 \text{ nF}$ va $R = 100 \text{ k}\Omega$. K kalit ulangan holda zanjir o'zgaruvchan tok manbaiga ulangan. Manba berayotgan tokning kuchi o'zgarmas, tok chastotasi esa o'zgartirilishi mumkin. a. Zanjirdagi aktiv quvvat maksimal P_m qiymatga erishadigan f_m chastota va aktiv quvvat maksimal quvvatning yarmiga teng bo'ladigan f_+ va f_- chastotalar farqi $\Delta f = f_+ - f_-$ nisbatini toping.



Elektr zanjiri sxemasi

Kalit K endi ochiq. Kalit ochilgandan so'ng darhol t_0 momentda L_1 va L_2 g'altaklaridagi tok kuchlari mos ravishda $i_{01} = 0.1 \text{ A}$ va $i_{02} = 0.2 \text{ A}$ (toklar rasmdagi yo'nalishda oqadi); shu momentda C_1 sig'imli kondensatordagi potentsiallar farqi $u_0 = 40 \text{ V}$: b. $L_1 C_1 C_2 L_2$ zanjiridagi elektromagnit tebranishlar chastotasini hisoblang; c. AB o'tkazgichdagi elektr tok kuchini aniqlang; d. L_1 g'altagidagi elektr tok kuchi tebranishining amplitudasini hisoblang.

G'altaklarning o'zaro induksiyasi va o'tkazgichlarning elektr qarshiligi hisobga olinmasin. Kalit ulanganda yoki uzilganda sodir bo'ladigan tez o'tish jarayonlari e'tiborga olinmasin.

II masala – Yechim

a. O'zgaruvchan tok zanjirlarini kompleks sonlar formalizmi yordamida o'rganishda ma'lumki, induktivlik L ga – chastotasi ω bo'lgan o'zgaruvchan tok bilan ta'minlangan zanjir qismiga – kompleks induktiv reaktiv qarshilik $X_L = L \cdot \omega \cdot j$, ($j = \sqrt{-1}$) mos keladi.

Shunga o'xshash, sig'im C ga kompleks sig'im reaktiv qarshiligi $X_C = \frac{1}{C \cdot \omega \cdot j}$ mos keladi.

Parallel zanjir uning kompleks o'tkazuvchanligi Y bilan tavsiflanadi. Rasmda ko'rsatilgan o'zgaruvchan tok zanjirining o'tkazuvchanligi:

$$\underline{Y} = \frac{1}{R} + \frac{1}{L_1 \cdot \omega \cdot j} + \frac{1}{L_2 \cdot \omega \cdot j} + C_1 \cdot \omega \cdot j + C_2 \cdot \omega \cdot j$$

$$\underline{Y} = \frac{1}{R} + j \cdot \left[(C_1 + C_2) - \left(\frac{1}{L_1 \cdot \omega} + \frac{1}{L_2 \cdot \omega} \right) \right] \quad (2.1)$$

Zanjir o'zini parallel ekvivalent sig'im C ga

$$C = C_1 + C_2 \quad (2.2)$$

va parallel ekvivalent induktivlik L ga

$$\frac{1}{L} = \frac{1}{L_1} + \frac{1}{L_2} \implies L = \frac{L_1 L_2}{L_1 + L_2} \quad (2.3)$$

ega bo'lgandek tutadi.

Zanjirning kompleks o'tkazuvchanligi quyidagicha yozilishi mumkin:

$$\underline{Y} = \frac{1}{R} + j \cdot \left(C \cdot \omega - \frac{1}{L \cdot \omega} \right) \quad (2.4)$$

va zanjirning kompleks qarshiligi:

$$\underline{Z} = \frac{1}{\underline{Y}} = \frac{1}{\frac{1}{R} + j \cdot \left(C \cdot \omega - \frac{1}{L \cdot \omega} \right)} \quad (2.5)$$

Zanjirning qarshiligi Z , ya'ni o'tkazuvchanlik Y ga teskari bo'lgan kattalik, kompleks qarshilik $\underline{Z}$ ning modulidir:

$$Z = |\underline{Z}| = \frac{1}{|\underline{Y}|} = \frac{1}{\sqrt{\frac{1}{R^2} + \left(C \cdot \omega - \frac{1}{L \cdot \omega} \right)^2}} \quad (2.6)$$

Zanjirni ta'minlovchi o'zgarmas tok manbai oniy qiymati $i(t)$ bo'lgan tok beradi:

$$i(t) = I \cdot \sqrt{2} \cdot \sin(\omega t) \quad (2.7)$$

bunda I tokning effektiv (o'zgarmas) kuchi, ω esa tokning siklik chastotasi (o'zgarishi mumkin). Zanjir qisqichlaridagi potentsiallar farqining oniy qiymati $u(t)$:

$$u(t) = U \cdot \sqrt{2} \cdot \sin(\omega t + \varphi) \quad (2.8)$$

bunda U kuchlanishning effektiv qiymati, φ esa kuchlanish va tok orasidagi faza farqi.

Tok va kuchlanishning effektiv qiymatlari quyidagi munosabatga bo'ysunadi:

$$U = I \cdot Z \quad (2.9)$$

Zanjirdagi aktiv quvvat:

$$P = \frac{U^2}{R} = \frac{Z^2 \cdot I^2}{R} \quad (2.10)$$

Shartga ko'ra,

$$\begin{cases} I = \text{o'zgarmas} \\ R = \text{o'zgarmas} \end{cases} \quad (2.11)$$

maksimal aktiv quvvat qarshilikning maksimal qiymatida, ya'ni o'tkazuvchanlikning minimal qiymatida amalga oshadi.

O'tkazuvchanlik

$$Y = \sqrt{\frac{1}{R^2} + \left(C \cdot \omega - \frac{1}{L \cdot \omega}\right)^2} \quad (2.12)$$

siklik chastota ω funksiyasi sifatida quyidagi eng kichik qiymatga ega:

$$Y_{min} = \frac{1}{R} \quad (2.13)$$

bu ω_m chastotada sodir bo'ladi:

$$\omega_m = \frac{1}{\sqrt{L \cdot C}} \quad (2.14)$$

Bu holda

$$C \cdot \omega_m - \frac{1}{L \cdot \omega_m} = 0 \quad (2.15)$$

Demak, zanjirdagi maksimal aktiv quvvat:

$$P_m = R \cdot I^2 \quad (2.16)$$

va u manba beradigan o'zgaruvchan tokning f_m chastotasida ro'y beradi:

$$f_m = \frac{\omega_m}{2\pi} = \frac{1}{2\pi\sqrt{C \cdot L}} \quad (2.17)$$

Aktiv quvvat maksimal quvvatning yarmiga teng bo'lishi uchun:

$$P = \frac{1}{2}P_m$$

$$\frac{Z^2 \cdot I^2}{R} = \frac{1}{2}R \cdot I^2$$

$$\frac{Z^2}{R^2} = \frac{1}{2} \implies Z^2 = \frac{R^2}{2}$$

ya'ni

$$\frac{1}{\frac{1}{R^2} + \left(C \cdot \omega - \frac{1}{L \cdot \omega}\right)^2} = \frac{R^2}{2}$$

$$\frac{1}{R^2} + \left(C \cdot \omega - \frac{1}{L \cdot \omega}\right)^2 = \frac{2}{R^2}$$

$$\left(C \cdot \omega - \frac{1}{L \cdot \omega}\right)^2 = \frac{1}{R^2} \quad (2.19)$$

Aktiv quvvatni maksimal quvvatning yarmiga teng qiladigan tok siklik chastotasi quyidagi tenglamalardan birini qanoatlantirishi kerak:

$$C \cdot \omega - \frac{1}{L \cdot \omega} = \pm \frac{1}{R}$$

$$\omega^2 \pm \frac{1}{R \cdot C} \omega - \frac{1}{L \cdot C} = 0 \quad (2.20)$$

Ikki kvadrat tenglama to'rtta yechim berishi mumkin:

$$\omega = \pm \left[\frac{1}{2R \cdot C} \pm \sqrt{\frac{1}{(2R \cdot C)^2} + \frac{1}{L \cdot C}} \right] \quad (2.21)$$

Siklik chastota har doim musbat bo'lgani va

$$\frac{1}{(2R \cdot C)^2} + \frac{1}{L \cdot C} > \frac{1}{(2R \cdot C)^2} \quad (2.22)$$

bo'lgani uchun faqat ikkita haqiqiy yechim mavjud:

$$\omega_{\pm} = \frac{1}{2R \cdot C} \pm \sqrt{\frac{1}{(2R \cdot C)^2} + \frac{1}{L \cdot C}} \quad (2.23)$$

Zanjirda aktiv quvvat maksimal quvvatning yarmini tashkil etadigan ikkita chastota mavjud $f_{\pm} = \frac{\omega_{\pm}}{2\pi}$:

$$\begin{cases} f_+ = \frac{1}{2\pi} \left[\frac{1}{2R \cdot C} + \sqrt{\frac{1}{(2R \cdot C)^2} + \frac{1}{L \cdot C}} \right] \\ f_- = \frac{1}{2\pi} \left[-\frac{1}{2R \cdot C} + \sqrt{\frac{1}{(2R \cdot C)^2} + \frac{1}{L \cdot C}} \right] \end{cases} \quad (2.24)$$

Bu chastotalar farqi:

$$\Delta f = f_+ - f_- = \frac{1}{2\pi R \cdot C} \quad (2.25)$$

bu zanjirning o'tkazish polosasi kengligi – rezonans chastota atrofidagi chastotalar oralig'i bo'lib, uning chekkalarida signal rezonans signalidan $1/\sqrt{2}$ marta kichik bo'ladi. O'tkazish polosasi chekkalarida aktiv quvvat rezonansdagi qiymatining yarmigacha kamayadi.

So'ralgan nisbat:

$$\frac{f_m}{\Delta f} = \frac{R \cdot C}{\sqrt{L \cdot C}} = R \sqrt{\frac{C}{L}}$$

$$\frac{f_m}{\Delta f} = R \sqrt{\frac{(C_1 + C_2) \cdot (L_1 + L_2)}{L_1 L_2}}$$

$$\frac{f_m}{\Delta f} = R \sqrt{\frac{C}{L}} = R \frac{\sqrt{(C_1 + C_2)(L_1 + L_2)}}{\sqrt{L_1 L_2}}$$

Berilgan qiymatlar: $C = 15 \text{ nF}$ $L = \frac{20}{3} \text{ mH}$ demak $\omega_m = 10^5 \text{ rad} \cdot \text{s}^{-1}$ va

$$\frac{f_m}{\Delta f} = R \sqrt{\frac{C}{L}} = 100 \times 10^3 \sqrt{\frac{3.15 \times 10^{-9}}{20 \times 10^{-3}}} = 150 \quad (2.27)$$

(2.26) munosabat a savolga javobdir.

- b. Manba uzilgandan so'ng darhol g'altaklarda tok mavjudligi, vaqtga bog'liq holda g'altaklardan tok o'tishda davom etishini taxmin qilishga imkon beradi.

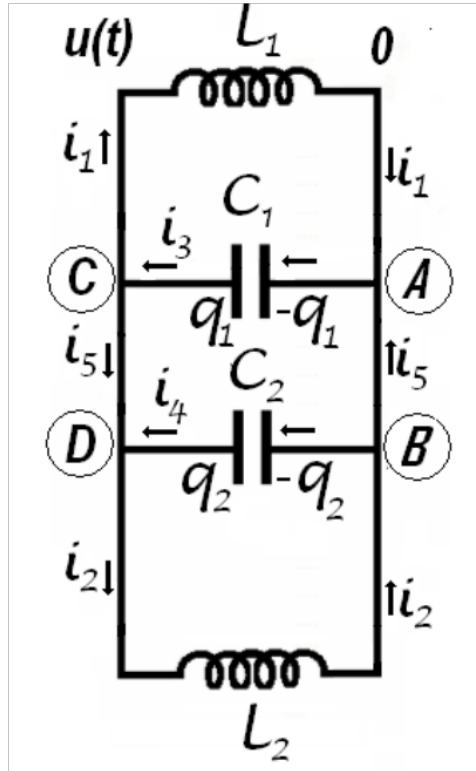


Figure 11: 2.1-rasm

Kondensatorlar vaqt bo'yicha o'zgaruvchi zaryadlar bilan zaryadlanadi. Kondensatorlar zaryadlarining o'zgarishi zanjirdagi kondensatorlarni bog'lovchi o'tkazgichlarda toklar paydo bo'lishiga olib keladi. G'altaklar va kondensatorlar qisqichlaridagi oniy kuchlanish – zanjirdagi barcha elementlar uchun bir xil – ham vaqtga bog'liqdir. Faraz qilaylik, C va D nuqtalarning elektr potentsiali $u(t)$ ga, A va B nuqtalarning potentsiali esa nolga teng. Agar L_1 induktivlikdan oniy qiymati $i_1(t)$ bo'lgan o'zgaruvchan tok o'tsa, tok va potentsiallar orasidagi munosabat:

$$u(t) - L_1 \frac{di_1}{dt} = 0 \quad (2.28)$$

Ikkinchi induktivlikdan o'tuvchi tok $i_2(t)$ uchun:

$$u(t) - L_2 \frac{di_2}{dt} = 0 \quad (2.29)$$

Agar C_1 sigʻimli kondensatorning musbat qoplamasida $q_1(t)$ zaryad toʻplangan boʻlsa, u holda kondensator qisqichlaridagi elektr kuchlanish $u(t)$ ga teng va

$$q_1 = C_1 \cdot u \quad (2.30)$$

Bu munosabatni differensiallab:

$$\frac{dq_1}{dt} = C_1 \frac{du}{dt} \quad (2.31)$$

Ammo

$$\frac{dq_1}{dt} = -i_3 \quad (2.32)$$

chunki elektr toki kondensator qoplamasidagi elektr zaryadining kamayishi hisobiga paydo boʻladi. Shuning uchun:

$$i_3 = -C_1 \frac{du}{dt} \quad (2.33)$$

Xuddi shunday, ikkinchi kondensator uchun:

$$i_4 = -C_2 \frac{du}{dt} \quad (2.34)$$

Olingan barcha natijalarni hisobga olgan holda:

$$\begin{aligned} \frac{di_1}{dt} &= \frac{u}{L_1} \\ \frac{di_2}{dt} &= \frac{u}{L_2} \end{aligned} \quad (2.35)$$

mos ravishda

$$\begin{aligned} \frac{di_3}{dt} &= -C_1 \frac{d^2u}{dt^2} \\ \frac{di_4}{dt} &= -C_2 \frac{d^2u}{dt^2} \end{aligned} \quad (2.36)$$

$i_5(t)$ deb B nuqtadan A nuqtaga oʻtuvchi tokning oniy kuchini belgilasak, u holda C va D nuqtalar orqali ham xuddi shunday oniy kuchli tok oʻtadi. A nuqta uchun Kirxgofning toklar qoidasi:

$$i_1 + i_5 = i_3 \quad (2.37)$$

B nuqta uchun esa:

$$i_4 + i_5 = i_2 \quad (2.38)$$

(2.37) va (2.38) dan:

$$i_1 - i_3 = i_4 - i_2 \quad (2.39)$$

va differensiallab:

$$\frac{di_1}{dt} - \frac{di_3}{dt} = \frac{di_4}{dt} - \frac{di_2}{dt} \quad (2.40)$$

ya'ni

$$\frac{u}{L_1} - \left(-C_1 \frac{d^2 u}{dt^2} \right) = \left(-C_2 \frac{d^2 u}{dt^2} \right) - \frac{u}{L_2}$$

$$u \left(\frac{1}{L_1} + \frac{1}{L_2} \right) = -\frac{d^2 u}{dt^2} (C_1 + C_2) \quad (2.41)$$

Yuqorida kiritilgan belgilashlardan foydalanib:

$$\frac{u}{L} = -C \frac{d^2 u}{dt^2}$$

$$\ddot{u} + \frac{1}{LC} u = 0 \quad (2.42)$$

Kuchlanish yuqoridagi munosabatga bo'ysungani uchun u vaqt bo'yicha garmonik bog'liqlikka ega bo'lishi kerak:

$$u(t) = A \cdot \sin(\omega t + \delta) \quad (2.43)$$

Kuchlanishning siklik chastotasi:

$$\omega = \frac{1}{\sqrt{LC}} \quad (2.44)$$

(2.43) va (2.36) munosabatlarni hisobga olgan holda:

$$\begin{aligned} i_3 &= -C_1 \frac{d}{dt}(A \cdot \sin(\omega t + \delta)) = -C_1 \cdot A \cdot \omega \cdot \cos(\omega t + \delta) \\ i_4 &= -C_2 \frac{d}{dt}(A \cdot \sin(\omega t + \delta)) = -C_2 \cdot A \cdot \omega \cdot \cos(\omega t + \delta) \end{aligned} \quad (2.45)$$

va

$$\begin{aligned} \frac{di_1}{dt} &= \frac{u}{L_1} = \frac{1}{L_1} A \cdot \sin(\omega t + \delta) \\ \frac{di_2}{dt} &= \frac{u}{L_2} = \frac{1}{L_2} A \cdot \sin(\omega t + \delta) \end{aligned} \quad (2.46)$$

Bundan:

$$\begin{aligned} i_1 &= \frac{1}{L_1 \omega} \cdot A \cdot \cos(\omega t + \delta) + M \\ i_2 &= \frac{1}{L_2 \omega} \cdot A \cdot \cos(\omega t + \delta) + N \end{aligned} \quad (2.47)$$

Yuqoridagi ifodalarda A, M, N va δ boshlang'ich shartlar yordamida aniqlanishi kerak bo'lgan doimiylardir. Kondensatorlardan o'tuvchi toklar sinusoidal, g'altaklardan o'tuvchi toklar esa sinusoidal va o'zgarmas toklarning yig'indisidan iborat ekanligi e'tiborga loyiq.

Birinchi momentda:

$$\begin{aligned} u(0) &= u_0 = 40\text{V} \\ i_1(0) &= i_{01} = 0.1\text{A} \\ i_2(0) &= i_{02} = 0.2\text{A} \end{aligned} \quad (2.48)$$

Induktivliklar va sig'implarning qiymatlari:

$$\begin{aligned} L_1 &= 0.01\text{H} \\ L_2 &= 0.02\text{H} \\ C_1 &= 10\text{nF} \\ C_2 &= 5\text{nF} \end{aligned} \quad (2.49)$$

Ekvivalent induktivlik va sig'im:

$$\begin{aligned} \frac{1}{L} &= \frac{1}{L_1} + \frac{1}{L_2} \\ L &= \frac{L_1 L_2}{L_1 + L_2} = \frac{2 \times 10^{-4}}{3 \times 10^{-2}} \text{H} = \frac{1}{150} \text{H} \end{aligned} \quad (2.50)$$

mos ravishda

$$\begin{aligned} C &= C_1 + C_2 \\ C &= 15\text{nF} \end{aligned} \quad (2.51)$$

(2.44) dan:

$$\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{\frac{1}{150} \cdot 15 \times 10^{-9}}} = 10^5 \text{ rad} \cdot \text{s}^{-1} \quad (2.52)^*$$

Siklik chastotaning qiymati soʻralgan b chastotani hisoblashga imkon beradi. Bu chastota f :

$$f = \frac{\omega}{2\pi} = \frac{10^5}{2\pi} \text{ Hz} \quad (2.53)^*$$

c. Agar zanjirdagi oniy kuchlanish (2.43) dagidek boʻlsa, quyidagilarni yozish mumkin:

$$\begin{cases} u(0) = A \cdot \sin(\delta) = u_0 \\ \sin(\delta) = \frac{u_0}{A} \end{cases} \quad (2.54)$$

(2.47) toklardan:

$$\begin{cases} i_{01} = \frac{1}{L_1 \cdot \omega} \cdot A \cdot \cos(\delta) + M \\ i_{02} = \frac{1}{L_2 \cdot \omega} \cdot A \cdot \cos(\delta) + N \end{cases} \quad (2.55)$$

Boshqa tomondan (2.39) ni quyidagicha ifodalash mumkin:

$$\frac{1}{L_1 \cdot \omega} \cdot A \cdot \cos(\omega \cdot t + \delta) + M + C_1 \cdot A \cdot \omega \cdot \cos(\omega \cdot t + \delta) = -C_2 \cdot A \cdot \omega \cdot \cos(\omega \cdot t + \delta) - \frac{1}{L_2 \cdot \omega} \cdot A \cdot \cos(\omega \cdot t + \delta) - N \quad (2.56)$$

$A \cos \alpha + B = C \cdot \cos \alpha + D$ (2.57) koʻrinishidagi ayniyat α argumentning istalgan qiymatida faqat

$$\begin{cases} A = C \\ B = D \end{cases} \quad (2.58)$$

boʻlgandagina oʻrinli boʻladi.

(2.58) ni hisobga olgan holda (2.56) dan:

$$\begin{cases} M + N = 0 \\ A \cdot \omega \cdot (C_1 + C_2) = -A \cdot \left(\frac{1}{\omega L_1} + \frac{1}{\omega L_2} \right) \end{cases} \quad (2.59)$$

Oxirgi tenglamadan zanjir (2.44) munosabatdagi siklik chastota bilan tebranishi kelib chiqadi. (2.55) munosabatlarni qo'shib, (2.54) va (2.59) ni hisobga olgan holda:

$$\begin{cases} i_{01} + i_{02} = A \cos(\delta) \cdot \left(\frac{1}{\omega L_1} + \frac{1}{\omega L_2} \right) \\ A = \frac{i_{01} + i_{02}}{\cos(\delta) \cdot \left(\frac{1}{\omega L_1} + \frac{1}{\omega L_2} \right)} \\ \cos \delta = \frac{i_{01} + i_{02}}{A \cdot \left(\frac{1}{\omega L_1} + \frac{1}{\omega L_2} \right)} \\ \cos \delta = \frac{(i_{01} + i_{02}) \cdot L \cdot \omega}{A} \end{cases} \quad (2.60)$$

Elektr kuchlanish amplitudasining son qiymati (2.54) va (2.60) dagi oxirgi munosabatlarni qo'shish orqali topiladi:

$$\begin{cases} \sin(\delta) = \frac{u_0}{A} \\ \cos \delta = \frac{(i_{01} + i_{02}) \cdot L \cdot \omega}{A} \\ (\cos(\delta))^2 + (\sin(\delta))^2 = 1 \\ \left(\frac{u_0}{A} \right)^2 + \left(\frac{(i_{01} + i_{02}) \cdot L \cdot \omega}{A} \right)^2 = 1 \\ A = \sqrt{(u_0)^2 + ((i_{01} + i_{02}) \cdot L \cdot \omega)^2} \end{cases} \quad (2.61)$$

Zanjir qisqichlaridagi elektr kuchlanishning son qiymati:

$$A = \sqrt{(40)^2 + \left((0.3) \cdot \frac{1}{150} \cdot 10^5 \right)^2} = \sqrt{(40)^2 + (200)^2} = 40\sqrt{26} \text{ V} \quad (2.62)$$

Va natijada (2.54) dan:

$$\begin{cases} \sin(\delta) = \frac{u_0}{A} \\ \sin(\delta) = \frac{40}{40\sqrt{26}} = \frac{1}{\sqrt{26}} \end{cases} \quad (2.63)$$

va

$$\cos(\delta) = \frac{5}{\sqrt{26}} \quad (2.64)$$

Shuningdek,

$$\begin{cases} \tan(\delta) = \frac{1}{5} \\ \delta = \arctan(1/5) \end{cases} \quad (2.65)$$

(2.55) dan:

$$\begin{cases} M = i_{01} - \frac{1}{L_1 \cdot \omega} \cdot A \cdot \cos(\delta) \\ N = i_{02} - \frac{1}{L_2 \cdot \omega} \cdot A \cdot \cos(\delta) \end{cases} \quad (2.66)$$

mos son qiymatlar:

$$\begin{cases} M = \left(0.1 - \frac{1}{0.01 \cdot 10^5} \cdot 40\sqrt{26} \cdot \frac{5}{\sqrt{26}} \right) \text{ A} = -0.1 \text{ A} \\ N = \left(0.2 - \frac{1}{0.02 \cdot 10^5} \cdot 40\sqrt{26} \cdot \frac{5}{\sqrt{26}} \right) \text{ A} = 0.1 \text{ A} \end{cases} \quad (2.67)^*$$

(2.47) munosabatlar quyidagi ko'rinishga keladi:

$$\begin{cases} i_1 = \frac{4\sqrt{26}}{100} \cdot \cos(10^5 \cdot t + \arctan(1/5)) - 0.1 \text{ A} = \tilde{i}_1 - I_0 \\ i_2 = \frac{2\sqrt{26}}{100} \cdot \cos(10^5 \cdot t + \arctan(1/5)) + 0.1 \text{ A} = \tilde{i}_2 + I_0 \end{cases} \quad (2.68)$$

G'altaklardan o'tuvchi toklar har xil amplitudali sinusoidal toklar va faqat g'altaklardan o'tuvchi o'zgarmas tokning superpozitsiyasidir. Bu o'zgarmas tokning qiymati:

$I_0 = 0.1 \text{ A}$ (2.69) 2.2-rasmdagidek.

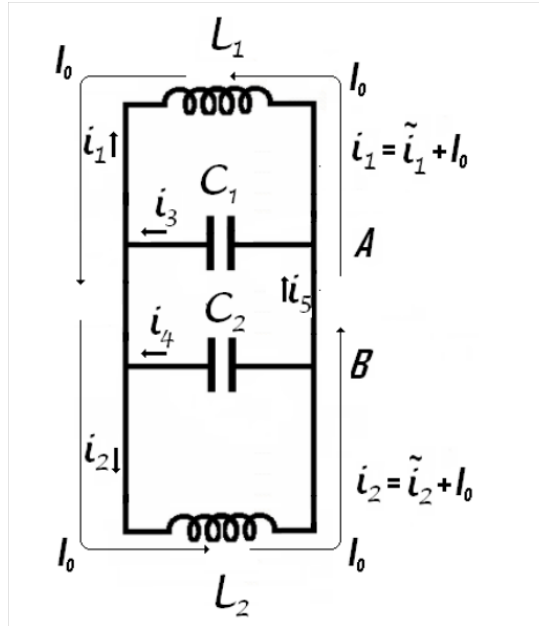


Figure 12: 2.2-rasm

G'altaklardan o'tuvchi o'zgaruvchan toklar quyidagi ifodalarga ega:

$$\begin{cases} \tilde{i}_1 = \frac{4\sqrt{26}}{100} \cdot \cos(10^5 \cdot t + \arctan(1/5)) \text{ A} \\ \tilde{i}_2 = \frac{2\sqrt{26}}{100} \cdot \cos(10^5 \cdot t + \arctan(1/5)) \text{ A} \end{cases} \quad (2.70)$$

Kondensatorlardan o'tuvchi toklar:

$$\begin{cases} i_3 = (-10 \times 10^{-9} \cdot 40\sqrt{26} \cdot 10^5 \cdot \cos(10^5 \cdot t + \arctan(1/5)))A \\ i_3 = \frac{4\sqrt{26}}{100} \cdot \cos(10^5 \cdot t + \arctan(1/5)) A \\ i_4 = (-5 \times 10^{-9} \cdot 40\sqrt{26} \cdot 10^5 \cdot \cos(10^5 \cdot t + \arctan(1/5)))A \\ i_4 = \frac{2\sqrt{26}}{100} \cdot \cos(10^5 \cdot t + \arctan(1/5)) A \end{cases} \quad (2.71)$$

i_5 tokning ifodasi:

$$\begin{cases} i_5 = i_3 - i_1 \\ i_5 = -\frac{8\sqrt{26}}{100} \cdot \cos(10^5 \cdot t + \arctan(1/5)) + 0.1 A \end{cases} \quad (2.72)$$

i_5 tok kuchining qiymati c savolga javobdir. Bu tokning boshlang'ich qiymati:

$$i_5 = \left(-\frac{8\sqrt{26}}{100} \cdot \frac{5}{\sqrt{26}} + 0.1 \right) A = -0.3 A \quad (2.73)^*$$

d. L_1 induktivlikdan o'tuvchi tokning amplitudasi:

$$\max(\tilde{i}_1) = \max \left(\frac{4\sqrt{26}}{100} \cdot \cos(10^5 \cdot t + \arctan(1/5)) \right) A = \frac{4\sqrt{26}}{100} A \approx 0.2 A \quad (2.74)^*$$

bu d savolga javobdir.

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Optika – III masala (7 ball)

Prizmalar

Cho‘qqi burchaklari $A_1 = 60^\circ$ va $A_2 = 30^\circ$ bo‘lgan ikkita dispersiyalovchi prizma rasmda ko‘rsatilganidek yopishtirilgan ($\hat{C} = 90^\circ$). Prizmalarning sindirish ko‘rsatkichlarining to‘lqin uzunligiga bog‘liqligi quyidagi munosabatlar bilan berilgan:

$$n_1(\lambda) = a_1 + \frac{b_1}{\lambda^2}$$

$$n_2(\lambda) = a_2 + \frac{b_2}{\lambda^2}$$

bunda $a_1 = 1.1$, $b_1 = 1 \cdot 10^3 \text{ nm}^2$, $a_2 = 1.3$, $b_2 = 5 \cdot 10^4 \text{ nm}^2$.

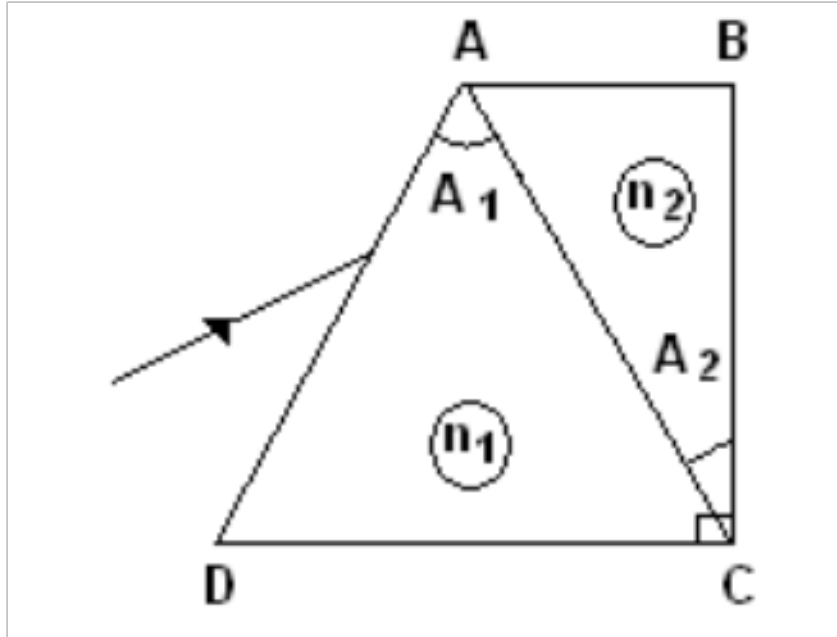


Figure 13: Prizmalar tizimi

- a. Ixtiyoriy tushish burchagida AC sirtida sinmasdan o‘tadigan nurlanishning λ_0 to‘lqin uzunligini aniqlang; prizmalarning mos sindirish ko‘rsatkichlarini toping. b. Tizimga bir xil burchak ostida tushuvchi uch xil nurlanish λ_{red} , λ_0 , λ_{violet} uchun prizmalar tizimidagi nurlar yo‘lini chizing. c. λ_0 to‘lqin uzunlikli nur uchun tizimdagi minimal og‘ish burchagini aniqlang. d. DC ga parallel yo‘nalishlar bo‘ylab tizimga kirib, undan chiqadigan nurning to‘lqin uzunligini hisoblang.

III masala – Yechim

- a. λ_0 to‘lqin uzunlikli nur, agar quyidagi shart bajarilsa, prizmalar tizimining AC sirtida ixtiyoriy tushish burchagida sinmasdan o‘tadi:

$$n_1(\lambda_0) = n_2(\lambda_0)$$

Prizmalar sindirish ko'rsatkichlarining to'liq uzunligiga bog'liqligi quyidagi ko'rinishda bo'lgani uchun:

$$n_1(\lambda) = a_1 + \frac{b_1}{\lambda^2} \quad (3.1)$$

$$n_2(\lambda) = a_2 + \frac{b_2}{\lambda^2} \quad (3.2)$$

(3.1) munosabat quyidagi ko'rinishga keladi:

$$a_1 + \frac{b_1}{\lambda_0^2} = a_2 + \frac{b_2}{\lambda_0^2} \quad (3.3)$$

λ_0 to'liq uzunligi mos ravishda quyidagicha ifodalanadi:

$$\lambda_0 = \sqrt{\frac{b_1 - b_2}{a_2 - a_1}} \quad (3.4)$$

Berilgan son qiymatlarni qo'yib:

$$\lambda_0 = 500 \text{ nm} \quad (3.5)$$

λ_0 to'liq uzunlikli nurlanish uchun prizmalar sindirish ko'rsatkichlarining umumiy qiymati:

$$n_1(\lambda_0) = n_2(\lambda_0) = 1.5 \quad (3.6)$$

(3.6) munosabatlar a savolga javobdir.

- b. Birinchi prizмага bir xil tushish burchagi bilan tushuvchi har xil to'liq uzunlikli nurlar (λ_{red} , λ_0 , λ_{violet}) uchun yo'llar 3.1-rasmda tasvirlangan.

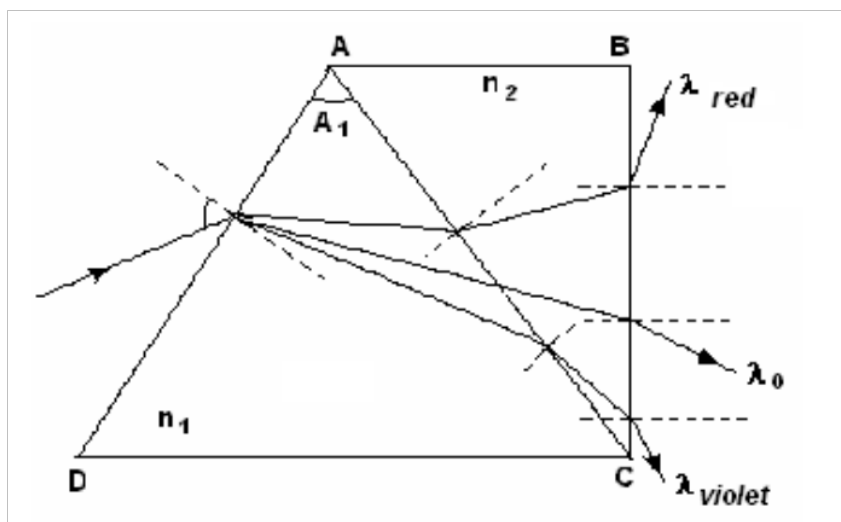


Figure 14: 3.1-rasm

3.1-rasmdagi chizma b savolga javobdir.

- c. 3.2-rasmda λ_0 to'liq uzunlikli nurning minimal og'ishdagi yo'li (nurning tushish yo'nalishi va chiquvchi nur yo'nalishi orasidagi burchak minimal bo'lgan holat) keltirilgan.

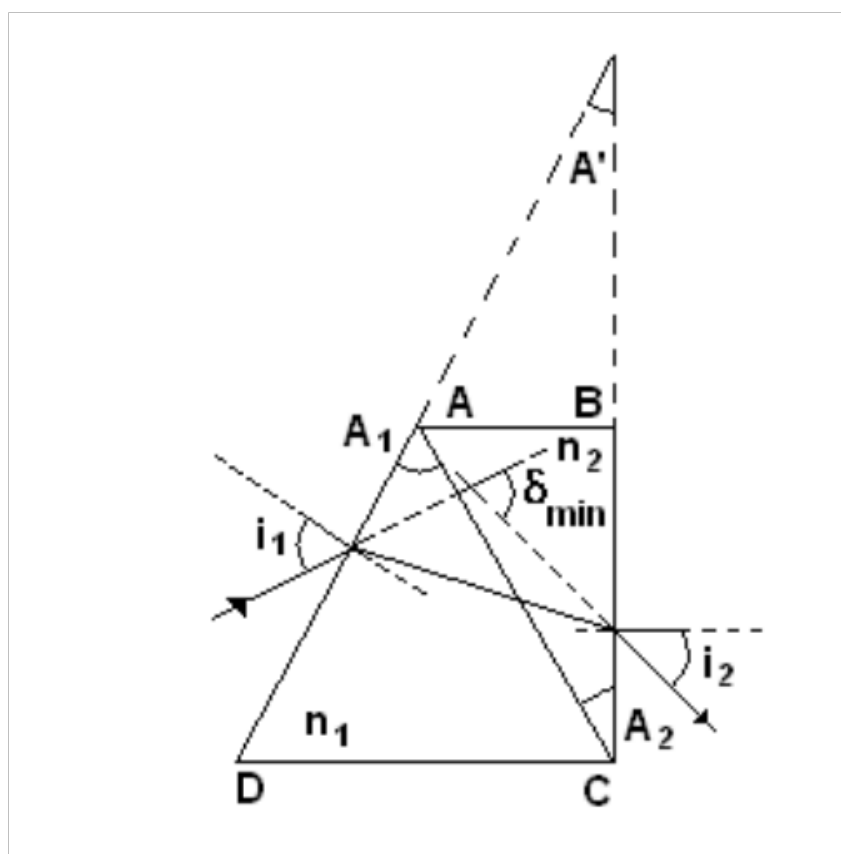


Figure 15: 3.2-rasm

Bu holatda

$$n_1(\lambda_0) = n_2(\lambda_0) = \frac{\sin\left(\frac{\delta_{\min} + A'}{2}\right)}{\sin\left(\frac{A'}{2}\right)} \quad (3.7)$$

bunda $m(A') = 30^\circ$, 3.1-rasmdagidek. (3.8) ga sindirish ko'rsatkichlarining qiymatlarini qo'yib:

$$\sin\left(\frac{\delta_{\min} + A'}{2}\right) = \frac{3}{2} \sin\left(\frac{A'}{2}\right) \quad (3.8)$$

yoki

$$\delta_{\min} = 2 \arcsin\left(\frac{3}{2} \sin\left(\frac{A'}{2}\right)\right) - A' \quad (3.9)$$

Son jihatdan $\delta_{\min} = 30.7^\circ$ (3.10) (3.11) munosabat c savolga javobdir.

d. 3.3-rasmdan foydalanib, AD sirtidagi sinish qonuni:

$$\sin i_1 = n_1 \sin r_1 \quad (3.11)$$

AC sirtidagi sinish qonuni:

$$n_1 \sin r'_1 = n_2 \sin r_2 \quad (3.12)$$

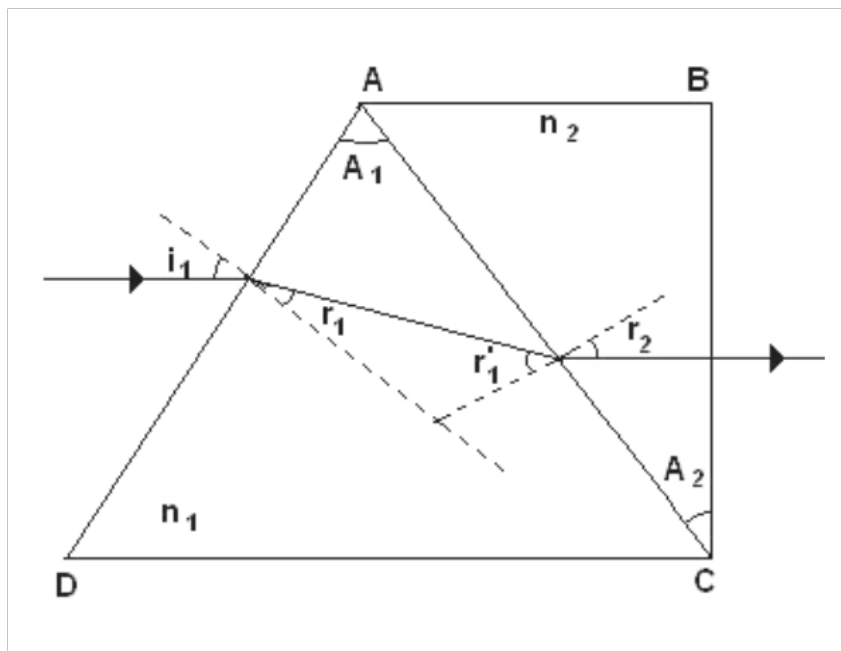


Figure 16: 3.3-rasm

3.3-rasmda ko‘rinib turibdiki, $r_2 = A_2$ (3.13) va $i_1 = 30^\circ$ (3.14) Shuningdek,
 $r_1 + r'_1 = A_1$ (3.15) (3.16) va (3.14) ni (3.13) ga qo‘yib:

$$\sqrt{4n_1^2 - 1} = \frac{2n_2 \sin A_2 + \cos A_1}{\sin A_1}$$

$A_1 = 60^\circ$ va $A_2 = 30^\circ$ bo‘lgani uchun (3.21) munosabat quyidagicha yoziladi:

$$\sqrt{4n_1^2 - 1} = \frac{2n_2 + 1}{\sqrt{3}} \quad (3.21)$$

yoki

$$3n_1^2 = 1 + n_2^2 + n_2^2 \quad (3.22)$$

(3.1), (3.2) va (3.23) munosabatlarni hisobga olib, barcha hisoblarni bajarib:

$$\lambda^4 \cdot (3a_1^2 - a_2^2 - a_2 - 1) + (6a_1b_1 - b_2 - 2a_2b_2) \cdot \lambda^2 + 3b_1^2 - b_2^2 = 0 \quad (3.23)$$

(3.24) tenglamani yechib, prizmalar tizimiga DC' ga parallel yo‘nalishda kirib, tizimdan yana DC ga parallel yo‘nalishda chiqadigan nurning λ to‘lqin uzunligi aniqlanadi. Ya’ni $\lambda = 1194 \text{ nm}$ (3.24) yoki $\lambda = 1.2 \mu\text{m}$ (3.25) (3.26) munosabat d savolga javobdir.

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Atom fizikasi – IV masala (7 ball)

Kompton sochilishi

λ_i to‘lqin uzunlikli foton harakatlanayotgan erkin elektron tomonidan sochiladi. Natijada elektron to‘xtaydi va hosil bo‘lgan λ to‘lqin uzunlikli foton tushuvchi foton yo‘nalishiga nisbatan $\theta = 60^\circ$ burchak ostida sochilib, yana ikkinchi tinch turgan erkin elektronga sochiladi. Bu ikkinchi sochilish jarayonida $\lambda_f = 1.25 \times 10^{-10}$ m to‘lqin uzunlikli foton λ to‘lqin uzunlikli foton yo‘nalishiga nisbatan $\theta = 60^\circ$ burchak ostida chiqadi. Birinchi elektronning o‘zaro ta’sirlashuvdan oldingi de Broyl to‘lqin uzunligini toping. Quyidagi doimiylar ma’lum:

$$h = 6.6 \times 10^{-34} \text{ J} \cdot \text{s} - \text{Plank doimiysi} \quad m = 9.1 \times 10^{-31} \text{ kg} - \text{elektron massasi} \\ c = 3.0 \times 10^8 \text{ m/s} - \text{yorug‘likning vakuumdagi tezligi}$$

IV masala – Yechim

Masalaning maqsadi birinchi elektronning tezligi, impulsi va to‘lqin uzunligini hisoblashdir.

Fotonlarni tavsiflash uchun quyidagi belgilashlar qo‘llaniladi: 4.1-jadval

boshlang‘ich foton	birinchi sochilishdan keyingi foton	oxirgi foton	—————
—————	—————	impulsi p_i p_0 p_f	energiyasi E_i E_0 E_f
		to‘lqin uzunligi λ_i λ_0 λ_f	

Elektronlarni tavsiflash uchun esa: 4.2-jadval

birinchi elektron to‘qnashuv oldidan	birinchi elektron to‘qnashuvdan keyin	ikkinchi elektron to‘qnashuv oldidan	ikkinchi elektron to‘qnashuvdan keyin	—————
—————	—————	—————	—————	
—————	impulsi p_{1e} p_{1e} 0 0 p_{2e}	energiyasi E_{1e} E_{1e} E_{0e} E_{0e}		
	E_{2e}	tezligi V_{1e} V_{1e} 0 0 V_{2e}		

4.1-rasmdagi tasvir fotonning birinchi sochilishidan oldingi holatni ko‘rsatadi.

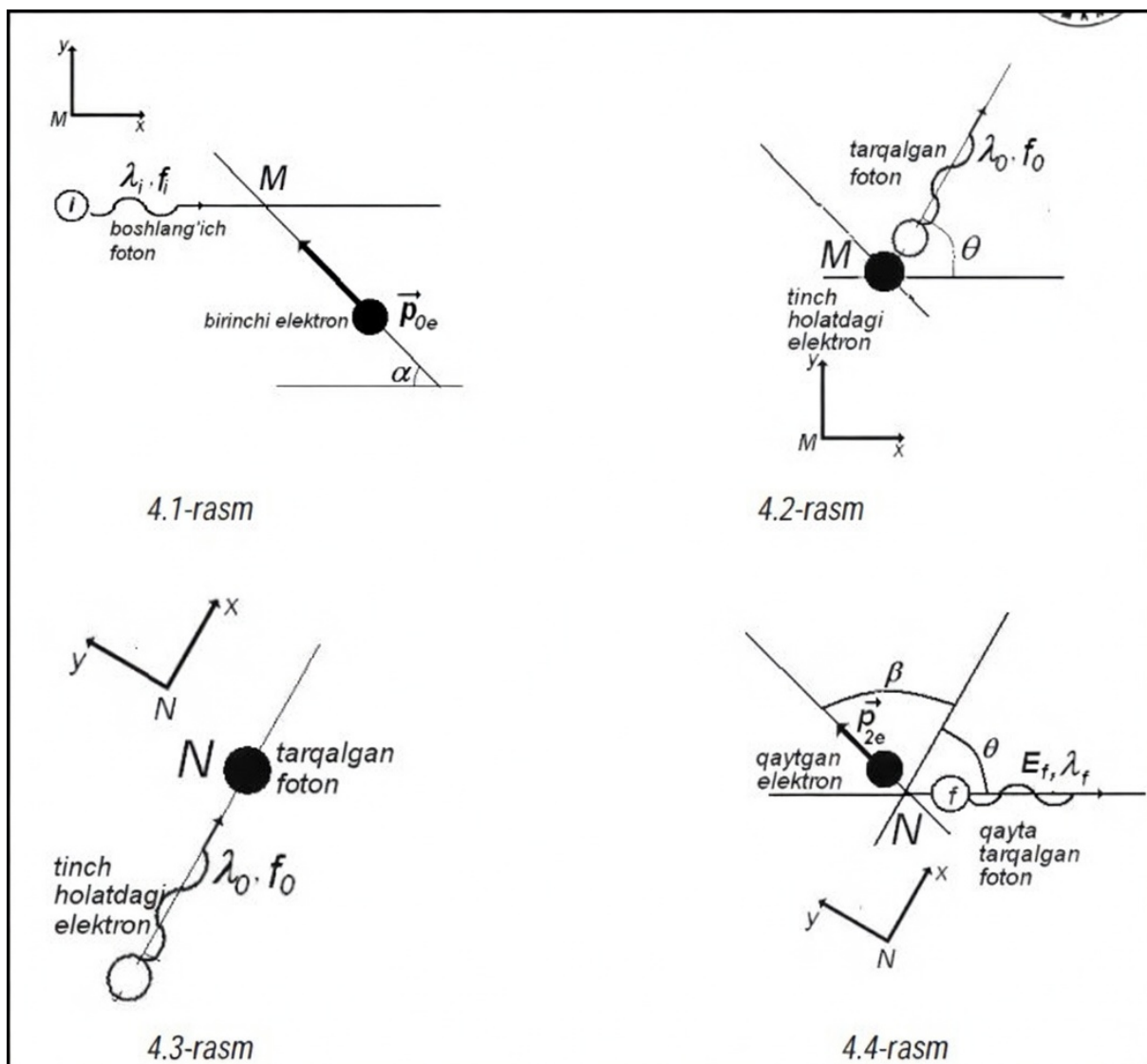


Figure 17: 4.1,2,3,4-rasmlar

Boshlang'ich fotonni tavsiflash uchun uning impulsi $\vec{p}_i$ va energiyasi E_i dan foydalanamiz.

$$\begin{cases} \vec{p}_i = \frac{h}{\lambda_i} = \frac{h \cdot f_i}{c} \\ E_i = h \cdot f_i \end{cases} \quad (4.1)$$

$$f_i = \frac{c}{\lambda_i} \quad (4.2)$$

boshlang'ich fotonning chastotasi. Boshlang'ich, harakatdagi erkin elektron uchun impuls $\vec{p}_{0e}$ va energiya E_{0e} :

$$\begin{cases} \vec{p}_{0e} = m \cdot \vec{V}_{1e} = \frac{m_0 \cdot \vec{V}_{1e}}{\sqrt{1-\beta^2}} \\ E_{0e} = m \cdot c^2 = \frac{m_0 \cdot c^2}{\sqrt{1-\beta^2}} \end{cases} \quad (4.3)$$

bunda m_0 elektronning tinchlikdagi massasi, m esa harakatlanayotgan elektronning massasi. Odatdagidek, $\beta = \frac{V_{1e}}{c}$. Birinchi elektronning de Broyl to'liqin uzunligi:

$$\lambda_{0e} = \frac{h}{p_{0e}} = \frac{h}{m_0 \cdot V_{1e}} \sqrt{1 - \beta^2}$$

Foton sochilishidan keyingi holat 4.2-rasmda tasvirlangan. Sochilgan fotonni tavsiflash uchun uning impulsi $\vec{p}_0$ va energiyasi E_0 dan foydalanamiz.

$$\begin{cases} \vec{p}_0 = \frac{h}{\lambda_0} = \frac{h \cdot f_0}{c} \\ E_0 = h \cdot f_0 \end{cases} \quad (4.4)$$

bunda

$$f_0 = \frac{c}{\lambda_0} \quad (4.5)$$

sochilgan fotonning chastotasi.

Sochilishdan keyin to'xtab qolgan elektronning impuls moduli nolga teng; uning energiyasi E_{1e} . To'qnashuvdan keyingi elektronning massasi m_0 – tinch turgan elektronning tinchlikdagi massasi. Demak, $E_{1e} = m_0 c^2$ Birinchi harakatlanayotgan elektronning impulsini aniqlash uchun impuls va energiyaning saqlanish qonunlarini yozish mumkin. Ya'ni $\vec{p}_i + \vec{p}_{0e} = \vec{p}_0$ (4.6) va $E_i + E_{0e} = E_0 + E_{1e}$ (4.7) Impulsning Ox o'qidagi saqlanishi:

$$\frac{h \cdot f_i}{c} + m \cdot V_{1e} \cos \alpha = \frac{h \cdot f_0}{c} \cos \theta \quad (4.8)$$

Impulsning Oy o'qidagi saqlanishi:

$$m \cdot V_{1e} \sin \alpha = \frac{h \cdot f_0}{c} \sin \theta \quad (4.9)$$

α ni yo'qotish uchun oxirgi ikki tenglamani quyidagicha yozamiz:

$$\begin{aligned} (m \cdot v_{1e} \cdot \cos \alpha)^2 &= \frac{h^2}{c^2} (f_0 \cos \theta - f_i)^2 \\ (m \cdot v_{1e} \cdot \sin \alpha)^2 &= \left(\frac{h \cdot f_0}{c} \sin \theta \right)^2 \end{aligned}$$

va keyin ularni qo'shamiz. Natija:

$$m^2 \cdot v_{1e}^2 = \frac{h^2}{c^2} (f_0^2 + f_i^2 - 2f_0 \cdot f_i \cdot \cos \theta)$$

yoki

Energiyaning saqlanishi (4.7) quyidagicha qayta yoziladi:

$$m \cdot c^2 + h \cdot f_i = m_0 \cdot c^2 + h \cdot f_0 \quad 4.13$$

yoki

$$\frac{m_0 \cdot c^2}{\sqrt{1 - \left(\frac{v_{1e}}{c}\right)^2}} = m_0 \cdot c^2 + h \cdot (f_0 - f_i)$$

Oxirgi munosabatni kvadratga ko'tarib:

$$\frac{m_0^2 \cdot c^4}{1 - \left(\frac{v_{1e}}{c}\right)^2} = m_0^2 \cdot c^4 + h^2(f_0 - f_i)^2 + m_0 \cdot h \cdot c^2 \cdot (f_0 - f_i)$$

(4.12) dan (4.15) ni ayirib:

$$2m_0 \cdot c^2 \cdot h \cdot (f_0 - f_i) + 2h^2 \cdot f_i \cdot f_0 \cdot \cos \theta - 2h^2 \cdot f_i \cdot f_0 = 0$$

yoki

$$\frac{h}{m_0 \cdot c}(1 - \cos \theta) = \frac{c}{f_i} - \frac{c}{f_0}$$

$$\Lambda = \frac{h}{m_0 \cdot c} \quad 4.18$$

belgilashdan foydalanib, (4.17) munosabat quyidagi ko'rinishga keladi:

$$\Lambda \cdot (1 - \cos \theta) = \lambda_f - \lambda_0 \quad 4.19$$

Sochilgan fotonning to'liq uzunligi:

$$\lambda_0 = \lambda_f - \Lambda \cdot (1 - \cos \theta) \quad 4.20$$

boshlang'ich foton to'liq uzunligidan qisqa, demak, sochilgan fotonning energiyasi boshlang'ich foton energiyasidan katta.

$$\begin{cases} \lambda_f < \lambda_0 \\ E_f > E_0 \end{cases}$$

Endi N nuqtada sodir bo'ladigan ikkinchi to'qnashuv jarayonini tahlil qilamiz. Buning uchun Ox yo'nalishi birinchi to'qnashuvdan keyin sochilgan foton yo'nalishi bo'ylab yo'naltirilgan yangi sanoq sistemasini qaraymiz.

4.3-rasm ikkinchi to'qnashuv oldidan holatni, 4.4-rasm esa bu sochilish jarayonidan keyingi holatni tasvirlaydi. Sochilish jarayonida impulsning saqlanish qonuni:

$$\begin{cases} \frac{h}{\lambda_0} \cos \theta + m \cdot v_{2e} \cdot \cos \beta = \frac{h}{\lambda_f} \\ \frac{h}{\lambda_0} \sin \theta - m \cdot v_{2e} \cdot \sin \beta = 0 \end{cases}$$

Noma'lum β burchakni yo'qotish uchun (4.22) tenglamalarni kvadratga ko'tarib qo'shish kerak. Ya'ni

$$\begin{cases} \left(\frac{h}{\lambda_0} \cos \theta - \frac{h}{\lambda_f} \right)^2 = (m \cdot v_{2e} \cos \beta)^2 \\ \left(\frac{h}{\lambda_0} \sin \theta \right)^2 = (m \cdot v_{2e} \sin \beta)^2 \end{cases}$$

yoki

$$\left(\frac{h}{\lambda_f} \right)^2 + \left(\frac{h}{\lambda_0} \right)^2 - \frac{2h^2}{\lambda_0 \cdot \lambda_f} \cos \theta = (m \cdot v_{2e})^2$$

Ikkinchi sochilish jarayonida energiyaning saqlanish qonuni:

$$\frac{h \cdot c}{\lambda_0} + m_0 \cdot c^2 = \frac{h \cdot c}{\lambda_f} + m \cdot c^2$$

(4.24) va (4.25) dan:

$$\frac{h^2 \cdot c^2}{\lambda_f^2} + \frac{h^2 \cdot c^2}{\lambda_0^2} - \frac{2h^2 \cdot c^2}{\lambda_0 \cdot \lambda_f} \cos \theta = m^2 \cdot c^2 \cdot v_{2e}^2$$

va

$$h^2 \cdot c^2 \left(\frac{1}{\lambda_f} - \frac{1}{\lambda_0} \right)^2 + m_0^2 \cdot c^4 + 2h \cdot c^3 \cdot m_0 \left(\frac{1}{\lambda_f} - \frac{1}{\lambda_0} \right) = m^2 \cdot c^4$$

(4.26) dan (4.27) ni ayirib:

$$\frac{h}{m_0 \cdot c} (1 - \cos \theta) = \lambda_f - \lambda_0$$

Ya'ni

$$\begin{cases} \lambda_f > \lambda_0 \\ E_f < E_0 \end{cases}$$

λ_f ning qiymati ma'lum va Λ ni hisoblash mumkin:

$$\begin{cases} \lambda_f = 1.25 \times 10^{-10} \text{ m} \\ \Lambda = \frac{6.6 \times 10^{-34}}{9.1 \times 10^{-31} \cdot 3 \times 10^8} \text{ m} = 2.41 \times 10^{-12} \text{ m} = 0.02 \times 10^{-10} \text{ m} \end{cases}$$

ikkinchi sohilish oldidan foton to'liq uzunligining qiymati:

$$\lambda_0 = 1.23 \times 10^{-10} \text{ m}$$

(4.28) ni quyidagicha yozib:

$$\lambda_f = \lambda_0 + \Lambda \cdot (1 - \cos \theta)$$

va (4.20) ni quyidagicha yozib:

$$\lambda_f = \lambda_0 + \Lambda \cdot (1 - \cos \theta)$$

aniq ko'rinadiki:

$$\lambda_f = \lambda_i$$

Ikki marta sohilgan fotonning energiyasi boshlang'ich foton energiyasi bilan bir xil. "Oxirgi foton" ning yo'nalishi "boshlang'ich" foton yo'nalishi bilan bir xil. Xulosa qilib aytganda, oxirgi foton boshlang'ich foton bilan aynan bir xil. Jarayonlarning simmetriyasi tufayli bu natija kutilgan edi. Simmetriya tahlilini elektronlarga qo'llasak, boshlang'ich fotonga to'qnashgan va undan keyin to'xtab qolgan birinchi harakatlanuvchi elektron, ikkinchi to'qnashuvdan keyingi ikkinchi elektron bilan bir xil impuls va energiyaga ega bo'lishi kerak – chunki bu ikkinchi elektron to'qnashuv oldidan tinch turgan edi. Ya'ni

$$\begin{cases} P_{1e} = P_{2e} \\ E_{1e} = E_{2e} \end{cases}$$

(4.24) ni hisobga olgan holda, oxirgi elektronning impulsi:

$$P_{2e} = h \sqrt{\frac{1}{\lambda_f^2} + \frac{1}{(\lambda_f - \Lambda(1 - \cos \theta))^2} - \frac{2 \cos \theta}{\lambda_f(\lambda_f - \Lambda(1 - \cos \theta))}}$$

Sochilishdan keyingi ikkinchi elektronning (va birinchi elektronning sochilish oldidan) de
Broyl to‘lqin uzunligi:

$$\lambda_{1e} = \lambda_{2e} = \frac{1}{\sqrt{\frac{1}{\lambda_f^2} + \frac{1}{(\lambda_f - \Lambda(1 - \cos \theta))^2} - \frac{2 \cos \theta}{\lambda_f(\lambda_f - \Lambda(1 - \cos \theta))}}}$$

Bu to‘lqin uzunligining son qiymati: $\lambda_{1e} = \lambda_{2e} = 1.24 \times 10^{-10} \text{ m}$

IPhO logotipi – V masala

Xalqaro Fizika Olimpiadasining logotipi quyidagi rasmda keltirilgan.

Rasmda suyuqlik oqimining silindrik sirt shakli atrofida egilishi hodisasi tasvirlangan. Suyuqlikning trayektoriyasi kutilgan uzuq chiziqli kabi emas, balki aylana shaklidagi yaxlit chiziqli kabi bo'ladi.

Bu hodisani (birinchi marta rumin muhandisi Genri Koanda tomonidan 1936 yilda kuzatilgan) sifat jihatidan tushuntirib bering.

Bu masala Olimpiadaning umumiy baliga kiritilmaydi. Eng yaxshi yechim maxsus mukofot bilan taqdirlanadi.



Figure 18: 5.1-rasm

V masala – Yechim

Faraz qilaylik, suyuqlik o'zgarmas bosimli idishda turibdi. Agar suyuqlikning (gaz yoki suyuqlik) kichik doiraviy yoki to'g'ri to'rtburchak kesimli ingichka oqimi soplodan muhitga chiqsa, muhitga tegishli zarralar oqim tomonidan olib ketiladi. Muhitga tegishli boshqa zarralar esa oqimga tortiladi.

Agar oqim katta sirt ustidan oqib o'tsa, oqim ustidagi muhit zarralari va oqim bilan sirt orasidagi zarralar oqim tomonidan olib ketiladi. Oqim ustidagi zarralar zichligi yangi kelayotgan zarralar hisobiga o'zgarmas qoladi, ammo sirt va oqim orasidagi zarralar o'rnini to'ldirib bo'lmaydi. Oqimning ustki va ostki tomonlari orasida bosim farqi paydo bo'lib, oqimni sirtga itaradi. Agar sirt egri bo'lsa, oqim uning shaklini takrorlaydi.

Quyidagi rasmdagi chap tasvir yuqori, o'zgarmas bosimli idishning soplosidan chiqayotgan suyuqlik oqimining normal oqishini ko'rsatadi. Suyuqlikning oxirgi bosimi muhit bosimiga teng. Rasmdagi o'ng tasvir esa suyuqlikning katta sirt ustidan oqishini ko'rsatadi. Oqim sirtga "yopishib" qoladi.

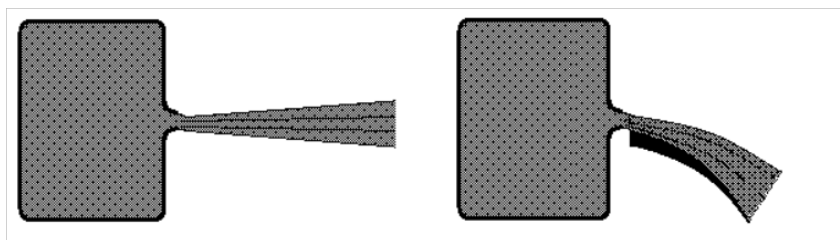


Figure 19: Koanda oqimi

Oqimning og'ish jarayoni oqim tezligini oshiradi, bunda oqimning bosimi va harorati o'zgarmaydi.

1936 yil dekabr oyida Parijda birinchi reaktiv samolyot sinovlari paytida rumin muhandisi Genri Koanda dvigatel alangalari qopqoqdan o'tayotganda yuzaga kelgan bu hodisani birinchi bo'lib kuzatgan.

Olimpiada logotipi suyuqlikning Koanda oqimini tasvirlaydi.